- Imperative commands (single line commands to create the objects)
  - **kubectl run my-pod --image=nginx** [here ***run*** is used, not ***create***]
    - **restartPolicy**: always
      always: restart pod even if the container exits with exit code ***0***
      onFailure: restart pod only if container's exit code is non-zero
      never: never restart the pod even if container stops.
    - **namespace**: default
      it is *logical* grouping of a cluster to separate logically like dev, prod, ..etc. You can set the resource allocation according to namespace as well.
    - **labels**: ***run****: my-pod*
    - port: not exposed
      [port is never exposed from container though, means from Service you can route to any port of the pod you want; *port* only there for ease of quering pods according to port or to display the usable port of the contaienr to user when they get *kubectl descripe pod <pod name>*]
    - (default configs of the command)

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: my-pod
  name: my-pod
spec:
  containers:
  - image: nginx
    name: my-pod
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
status: {}
```

  - **kubectl create deployment my-deploy --image=nginx** [here ***create*** is used]
    - Deployment name: my-deploy
    - ReplicaSet: my-deploy-<hash>
    - Pod: my-deploy-<hash>-<random>

- labels: ***app***:*my-deploy*
- These are the things kubectl create commands can create. pod cannot be created using kubectl create command

```
clusterrole           Create a cluster role
clusterrolebinding    Create a cluster role binding for a particular cluster role
configmap             Create a config map from a local file, directory or literal value
cronjob               Create a cron job with the specified name
deployment            Create a deployment with the specified name
ingress               Create an ingress with the specified name
job                   Create a job with the specified name
namespace             Create a namespace with the specified name
poddisruptionbudget   Create a pod disruption budget with the specified name
priorityclass         Create a priority class with the specified name
quota                 Create a quota with the specified name
role                  Create a role with single rule
rolebinding           Create a role binding for a particular role or cluster role
secret                Create a secret using a specified subcommand
service               Create a service using a specified subcommand
serviceaccount        Create a service account with the specified name
token                 Request a service account token
```

- 

(default configs of the command)

labels are automatically added.

- ***ReplicaSet*** cannot be created directly using imperative way, because Deployment is there to create and manage replicasets.
- ***kubectl expose*** takes replication controller, service, deployment, or pod and creates a *Service* to expose that.

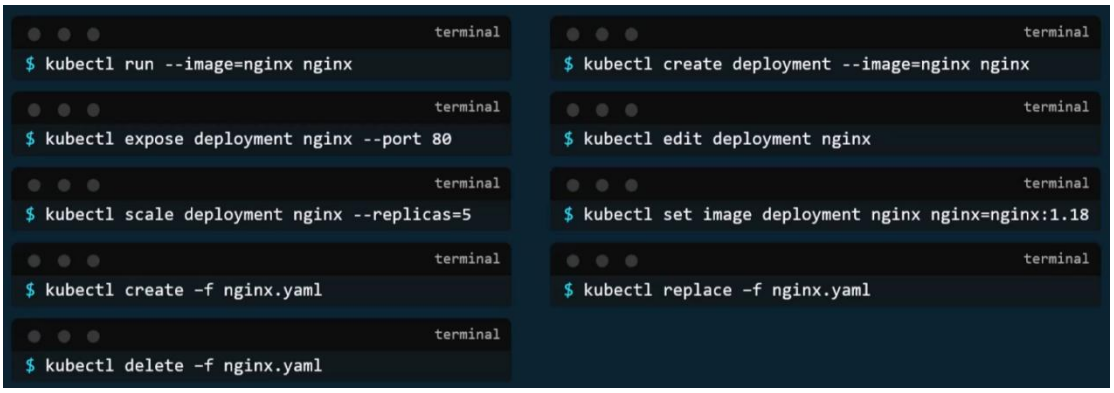
```
vagrant@controlplane:~$ kubectl get svc,pod
NAME                 TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.96.0.1     <none>        443/TCP   36m

NAME         READY   STATUS    RESTARTS   AGE
pod/my-pod   1/1     Running   0          34s
vagrant@controlplane:~$ kubectl expose pod my-pod --port 80
service/my-pod exposed
vagrant@controlplane:~$ kubectl get svc,pod
NAME                 TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.96.0.1      <none>        443/TCP   37m
service/my-pod       ClusterIP   10.96.68.232   <none>        80/TCP    6s

NAME         READY   STATUS    RESTARTS   AGE
pod/my-pod   1/1     Running   0          76s
```

- you can see here, It created a service named *my-pod*. Ifs of type ***ClusterIP*** which is default.
- If you give ***--type=NodePort*** flag, then it'll create a service of type NodePort

```
vagrant@controlplane:~$ kubectl expose pod my-pod --port 80 --type NodePort
service/my-pod exposed
vagrant@controlplane:~$ kubectl get svc,pod
NAME                 TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
service/kubernetes   ClusterIP   10.96.0.1       <none>        443/TCP        39m
service/my-pod       NodePort    10.96.119.232   <none>        80:32530/TCP   3s
```



[these are the imperative commands]

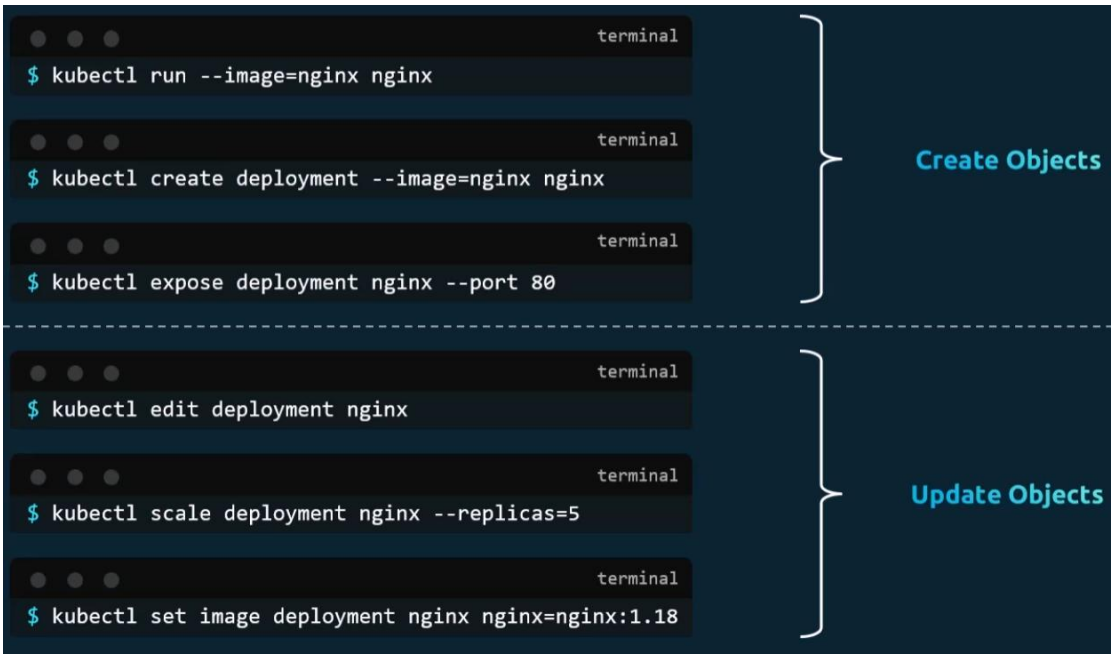
create -f is imperative because if already the same config present it'll fail, where as apply -f will not fail.

- Important trick
  - **kubectl create deployment my-deploy --image=nginx --dry-run=client -o yaml**

    kubectl run my-pod --image=nginx --dry-run=client -o yaml

- it'll give the yaml file of default deployment, remove the unnecessary fields, customize that, and create deployment
- Because of the flag ***--dry-run=client*** it'll not create the deployment, just simulate and give the output as yaml.

- **kubectl get pod my-pod -o yaml**
  - It'll give the running config of the pod as an yaml.
  - Can be done with any types of objects i.e. pod, replica set, container runtime, deployment ..etc
  - But it'll contain a lot of things, that needs to be removed.



- **edit** vs **replace** command
  - *kubectl edit* command open the running status of the object (pod, replicaset, replication controller, deployment, service ..etc) in a yaml file.

    This configurations are there in the memory, not stored in the yaml file that was being used to create the object.

    So, if you do ***kubectl edit*** then it'll not have any track of the changes. If someone later want to update the object, and he sees the yaml file present (not the running configuration yaml) and try to update, it might do some wrong changes in the running object.
  - *kubectl replace* command uses the physical yaml file to update the running object.

It does ***dumb*** update in the running object. Whatever there in the physical yaml, it'll update all the things and delete the configs which are not present in that physical yaml file.

By default it doesn't delete the current object (if anything doesn't match with physical yaml), to replace force fully you can give the flag ***--force***.

```
vagrant@controlplane:~$ kubectl replace --force -f pd.yaml
pod "my-pod" deleted from default namespace
pod/my-pod replaced
```

It looks similar to ***kubectl apply***, but *kubectl apply* does ***intelligent*** update. It finds the *diff* between running object and the physical yaml (used to update the object) and update those only.

It doesn't delete any configs which are not present in the current yaml but there in the running object.

kubectl apply is *declarative* command.

- If there are multiple files out of which objects will have to be created, then just give the path of the directory where those are present to *kubectl apply*, it'll create all those.

  kubectl apply -f <path where all files are present>

--- kubectl apply internals ---

- [here I'll take an example of a deployment named *my-dep*]
- There are 3 things.

  yaml (file that you write)

  last applied config (when you execute *kubectl* ***get*** *deploy my-dep* ***-o yaml***)

  [JSON format]

  live config (when you execute *kubectl* ***describe*** *deploy my-dep*)

  [*deploy* is shorthand of *deployment*]

  - Whenever you do *declarative* change using a yaml i.e.

    kubectl apply -f my-dep.yaml

    it changes both *live config*, *last applied config*. and obviously *yaml* is changed by you only.
  - Whenever you do *imperative* change using the command i.e.

    kubectl scale deployment my-dep --replicas=5

    it changes only *live config*. *Last applied config* will not be changed. And of cource, *yaml* has not been changed.

command type	yaml	last appplied config	live config
declarative	Y	Y	Y
imperative	N	N	Y

- There are a few commands [deployment is used for example; can be used for any objects]:
  - **kubectl diff -f <**yaml file name**>**

    compare the yaml with the *live config* along with last applied and display how it'll look like after you execute *kubectl apply*.
  - **kubectl apply view-last-updated deployment <**deployment name**>**

    show the last updated configs [which is present actually in JSON] in *yaml* format cleanly.
  - **kubectl get deployment <**deployment name**> -o yaml**

    kubectl get -f <deployment *yaml file* name**> -o yaml** [it fetch from the file]

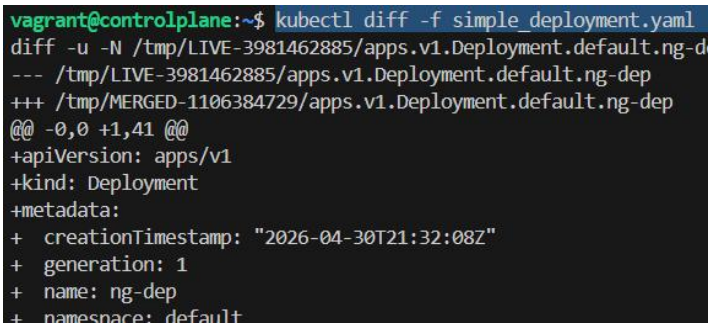
    Both will display the live configs of the deployment object.

    Also, there will be a JSON structure under *annotations* field where you can see *last updated config*.

- Stage-1: [create a deployment from yaml]
  - Initial yaml

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

  - **kubectl diff -f <**yaml file name**>** will display like below



[cropped]

  - Run the deployment using: *kubectl apply -f <yaml file name>*

- **kubectl apply view-last-applied deployment** <deployment name>

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  name: ng-dep
  namespace: default
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

- Stage 2: [change image using imperative command]
  - **kubectl set image deploy** <deployment name> <container name>=<new image>

    kubectl set image deploy ng-dep nginx=nginx:1.14.2

```
vagrant@controlplane:~$ kubectl set image deploy ng-dep nginx=nginx:1.14.2
deployment.apps/ng-dep image updated
```

  - **kubectl diff -f** <deployment file name>

```
vagrant@controlplane:~$ kubectl diff -f simple_deployment.yaml
diff -u -N /tmp/LIVE-2986659376/apps.v1.Deployment.default.ng-d
--- /tmp/LIVE-2986659376/apps.v1.Deployment.default.ng-dep
+++ /tmp/MERGED-1145231250/apps.v1.Deployment.default.ng-dep
@@ -6,7 +6,7 @@
     kubectl.kubernetes.io/last-applied-configuration: |
       {"apiVersion":"apps/v1","kind":"Deployment","metadata":{
ntainers":[{"image":"nginx:1.16.1","name":"nginx","ports":[{"co
   creationTimestamp: "2026-04-30T21:32:53Z"
-  generation: 2
+  generation: 3
   name: ng-dep
   namespace: default
   resourceVersion: "33813"
@@ -29,7 +29,7 @@
         app: nginx
     spec:
       containers:
-      - image: nginx:1.14.2
+      - image: nginx:1.16.1
         imagePullPolicy: IfNotPresent
```

    [you can see, yaml file has *nginx:1.16.1* but live config has *nginx:1.14.2*]

    [so, its showing `-` on *nginx:1.14.2* and `+` on *nginx:1.16.1*]

- **kubectl apply view-last-applied deploy <deployment name>**

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  name: ng-dep
  namespace: default
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

[its showing image as *nginx:1.16.1* because imperative change doesn't effect last applied config]

- Stage-3: [update yaml, remove replica field, then declarative update]

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

```
vagrant@controlplane:~$ kubectl diff -f simple_deployment.yaml
diff -u -N /tmp/LIVE-3508972230/apps.v1.Deployment.default.ng-
--- /tmp/LIVE-3508972230/apps.v1.Deployment.default.ng-dep
+++ /tmp/MERGED-2827279498/apps.v1.Deployment.default.ng-dep
@@ -6,14 +6,14 @@
     kubectl.kubernetes.io/last-applied-configuration: |
       {"apiVersion":"apps/v1","kind":"Deployment","metadata":
ntainers":[{"image":"nginx:1.16.1","name":"nginx","ports":[{"c
   creationTimestamp: "2026-04-30T21:32:53Z"
-  generation: 2
+  generation: 3
   name: ng-dep
   namespace: default
   resourceVersion: "33813"
   uid: 92189358-0dd5-4f8f-8144-1936af5322ab
 spec:
   progressDeadlineSeconds: 600
-  replicas: 3
+  replicas: 1
   revisionHistoryLimit: 10
   selector:
     matchLabels:
@@ -29,7 +29,7 @@
         spec:
           containers:
-        - image: nginx:1.14.2
+        - image: nginx:1.16.1
```

[its showing if I update, *replica* and *image* will be updated]

Now in last applied config, *replicas* field would have been removed because we deleted that field in yaml.

```
vagrant@controlplane:~$ kubectl apply view-last-applied -f simple_deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  name: ng-dep
  namespace: default
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

[replicas is not there, image is *nginx:1.16.1*]

- *live config*

```
spec:
  progressDeadlineSeconds: 600
  replicas: 1
```

```
spec:
  containers:
  - image: nginx:1.16.1
```

[the image and replicas got changed]

- Its looking like, why there is need of *last applied config* because when we apply, the live config is becoming same as the yaml.
- <u>Stage-1: [create declarative from yaml]</u>
  - See the yaml now

```
vagrant@controlplane:~$ cat simple_deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
  labels:
    app: ng-deployment
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:1.16.1
        ports:
        - containerPort: 80
```

[added one label only]

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  labels:
    app: ng-deployment
  name: ng-dep
  namespace: default
spec:
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - image: nginx:1.16.1
        name: nginx
        ports:
        - containerPort: 80
```

[last applied config will be same as yaml because of declarative change]

- Stage-2: [add one more label in imperative way; *kubectl edit* command]

```
labels:
  app: ng-deployment
  keyx: valx
```

[live config changed]

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  labels:
    app: ng-deployment
  name: ng-dep
```

[last applied config will be same because it was imperative update]

- Stage-3: [update the replicas using declarative]

```
vagrant@controlplane:~$ cat simple_deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
  labels:
    app: ng-deployment
spec:
  replicas: 3
  selector:
```

[updated yaml file]

```
    creationTimestamp: "2026
-   generation: 1
+   generation: 2
    labels:
      app: ng-deployment
      keyx: valx
@@ -16,7 +16,7 @@
    uid: 5071163c-87e2-49db-
  spec:
    progressDeadlineSeconds:
-   replicas: 1
+   replicas: 3
    revisionHistoryLimit: 10
```

[ignore generation]

[kubectl diff -f <deployment file name>]

you can see, its not showing *keys: valx* label to be deleted when it is not there in our current yaml file.

- kubectl apply -f <deployment file name>

```
vagrant@controlplane:~$ kubectl apply view-last-applied deploy ng-dep
apiVersion: apps/v1
kind: Deployment
metadata:
  annotations: {}
  labels:
    app: ng-deployment
  name: ng-dep
  namespace: default
spec:
  replicas: 3
```

[last applied config]

```
  creationTimestamp: "2026-04-
  generation: 2
  labels:
    app: ng-deployment
    keyx: valx
  name: ng-dep
  namespace: default
  resourceVersion: "37619"
  uid: 5071163c-87e2-49db-a1a0
spec:
  progressDeadlineSeconds: 600
  replicas: 3
```

[live config, replicas got increased but that label (keyx: valx) is still there]

- So, how does ***kubectl apply*** work then?
  - It does the below

    Sets fields, that appear in the configuration (yaml) file, in the live

configuration.

Clears fields, that removed from the configuration (yal) file, in the live configuration.

- It does the above after comparing with *last applied field*.
- If any field is there in [either of *yaml* or *last applied config*] (or) [the field is present in both *yaml* and *last applied config*]
  then that field will be updated in the *live config*.
- In the previous example. The label (keyx: valx) was not there in both yaml and *last applied config*, so it didn't track that.

In Simple Words it does the below

- It tracks the configs which are present in either yaml or last applied config and updates the live config accordingly. If something is there in live config but not there in both yaml and last applied config then it'll skip that.

Merging changes to primitive fields

Primitive fields are replaced or cleared.

Note: – is used for "not applicable" because the value is not used.

Field in object configuration file	Field in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Set live to configuration file value.
Yes	No	-	Set live to local configuration.
No	-	Yes	Clear from live configuration.
No	-	No	Do nothing. Keep live value.

Merging changes to map fields

Fields that represent maps are merged by comparing each of the subfields or elements of the map:

Note: – is used for "not applicable" because the value is not used.

Key in object configuration file	Key in live object configuration	Field in last-applied-configuration	Action
Yes	Yes	-	Compare sub fields values.
Yes	No	-	Set live to local configuration.
No	-	Yes	Delete from live configuration.
No	-	No	Do nothing. Keep live value.

yaml	last appplied config	live config	RES
Y	Y	Y/N	yaml : Y last : update to yaml live : update to yaml
Y	N	Y/N	yaml : Y last : update to yaml live : update to yaml
N	Y	Y/N	yaml : N last : remove live : remove
N	N	Y	yaml : N last : N live : no update

Namespaces

- Consider the below analogy

  - There are 2 houses,

    In one house (house-A), one person called Mark *Smith* is there.

    In another house (house-B), one person called Mark *Williams* is there.
  - In house-A, people can call *Mark Smith* as only *Mark* because there is no other guy having this name in that house.

    Similarly in house-B, people can call *Mark Williams* as *Mark* only.
  - But, if people of house-A wants to call *Mark Williams* then they need to call by full-name i.e. *Mark Williams*. Not only *Mark*.

    Similarly for people of house-B as well. They need to call *Mark Smith* .
  - A person who neither belong to house-A nor house-B, needs to call those by their full name only i.e. *Mark Smith* or *Mark Williams*.
  - Consider these houses as namespaces.

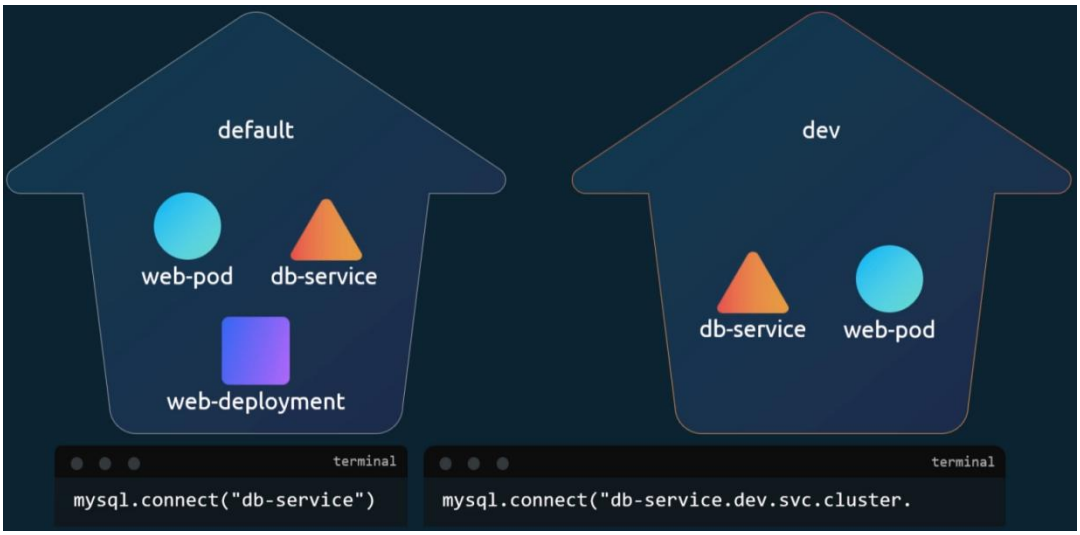
    Inside a namespace, resources having same names cannot be present (you can't create 2 pods having exactly same name). but, in multiple namespaces, resources of same name can be present.
  - For example, if you want multiple environments like *dev*, *prod* and you don't want to update any *production* pods while testing during *development* or something like that, ***namespaces*** help in that.

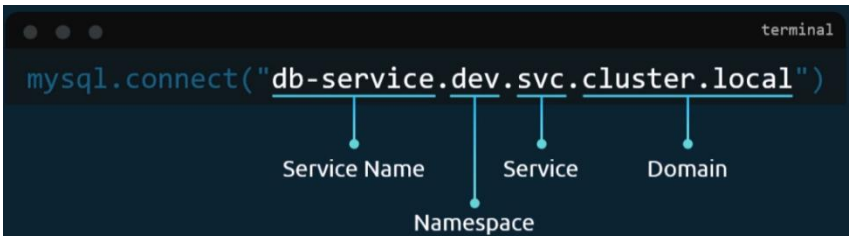
    You can have exact same resources in both the namespaces.

- Example:
  - Consider a single namespace scenario:
    - Some pods are running DB. And service name (ClusterIP) is ***db***.

- The application pods who want to use database can use ***db*** directly to access the database.

- Consider a multiple namespace scenario:
  - In the same example as above, lets imagine 2 namespaces are there.
  - Now, the application pods of one namespace can directly use ***db*** to access the database running in same namespace.
    But, if they want to use ***db*** of the other namespace then they need to give the full DNS path i.e. ***db.<namespace name>***
    or ***db.<namespace name>.svc.cluster.local*** [you don't need to give this, *db.<namespace name>* is converted to this internally]
- **db.<namespace name>.svc.cluster.local**
  in here, **svc** means don't access the pod directly, rather do using the ***db*** service.
  cluster.local is the internal DNS of the cluster. [**.local** is a top-level-domain just like *.com*, *.in* ..etc but it is internal]





- If you want to get or create the resources present in a specific namespace then
  kubectl get <resource type like *pod, rs, deploy*> **-n** <namespace name>
  kubectl create -f <yaml file name> **-n <namespace name>**

[**NOTE**: You need to write '***create***', not '***apply***' when you are creating something from the yaml first time (still have to verify this)]

[if the **namespace** key is present in the *yaml* file already, then no need to give ***--namespace*** flag]

- Or, in ***yaml*** you can write ***namespace*** inside metadata section.

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  namespace: dev
```

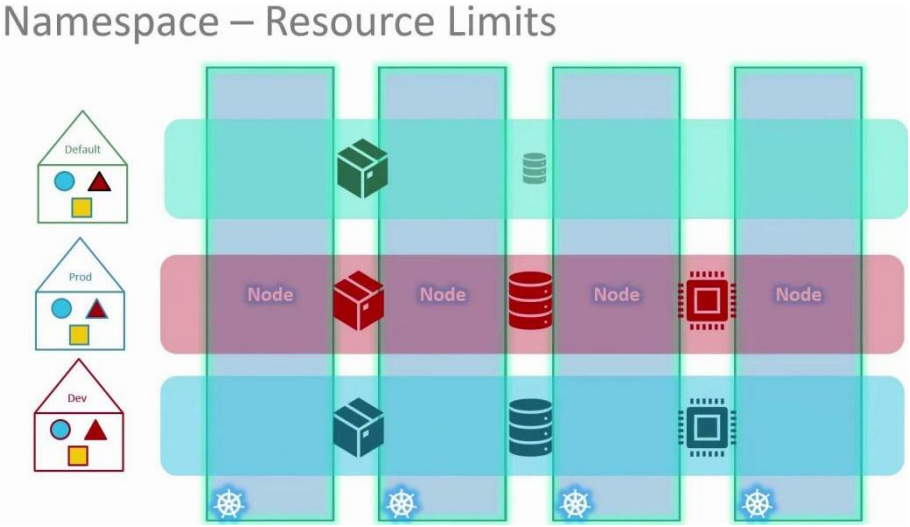
- To create a *namespace*, the yaml file format will be like this

```
vagrant@controlplane:~$ kubectl create namespace test --dry-run=client -o yaml
apiVersion: v1
kind: Namespace
metadata:
  name: test
```

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: dev
```

- All the ***namespace*** configures while setup:
  - When a kubernetes *cluster* is first setup, ***default*** namespace is created. Whatever resources you create, it is created inside this.
  - Kubernetes creates a set of Pods and Services for its internal puspose. These all stays in the namespace called ***kube-system*** .
  - There is another namespace called ***kube-public*** where resources available to all users are created.

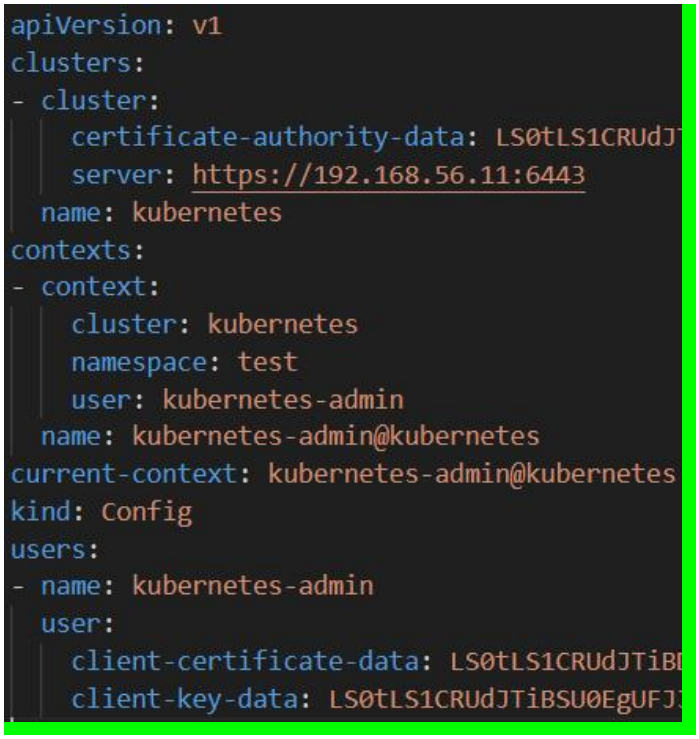
- You can create **quota** (long format: **ResourceQuota**) to the namespaces (setting the limits)

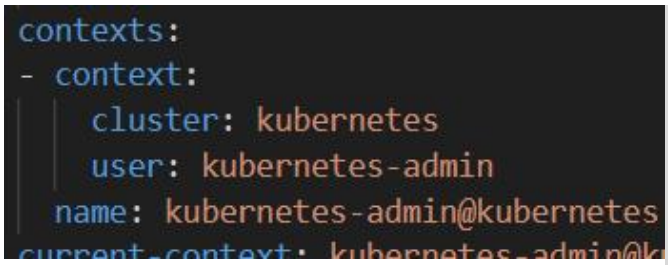


- If you want to change the default (not talking about the namespace having name *default*) namespace

  kubectl config set-context $(kubectl config current-context) --namespace=<namespae name which has to be made default**>**

kubeconfig

- The config file path is *~/.kube/config* .
- 

- 

[if namespace is not there]

  it looks something like this (its content format is of yaml, but the filename is *config* only, not ~~*config.yaml*~~)
- It mainly contains three things: **cluster, contexts, users**
- **clusters**
  - its a list of clusters.
  - Each cluster contains *name, cluster* (cluster contains *certificate-authority-data, server)* fields.
  - **name**
    - its just an identifier that will be used later inside *context* (you can see inside *context* field).
    - every operation i.e. create pod, get svc ..etc hits this URL.

      kubectl → API Server → Cluster

- *cluster.*server

  it's the end-point of ***API Server*** where all ***kubectl*** command goes.

- *cluster.*certificate-authority-data
  - Ensures you're talking to the real cluster
  - Prevents man-in-the-middle attacks

- users
  - It contains the list of users.
  - It defines "Who are you when talking to the Cluster"
  - Each entries of *user* contain the below fields **name, user** (*user* contains *client-certificate-data, client-key-data*) fields.
  - name

    just a identifier to the user. Which is referenced inside *context* field (you can see in the yaml).
  - user.client-certificate-data & user.client-key-data (common in local setup)

    certificate is like ID card*;* ***key*** is like signature (Analogy)
  - In real-world, instead of certificate and key, ***token*** is used.

    ```
    token: eyJhbGciOiJSUzI1NiIsIn
    ```

    - it is used in ServiceAccounts, CI/CD, API access.
  - In cloud provider cases, external Auth is present.
  - All-In-One, ***user*** field is for authentication with the ***cluster*** .

- contexts
  - It is something that reference all the things i.e.

    context = cluster + user + default namespace --- IMPORTANT ---
  - When you switch to a context, all the things will be switched i.e. user, cluster, *default* namespace
  - When you execute any command like

    kubectl get pod

    it is executed like the below

    kubectl get pod --namespace=<default namespace name>
  - If the namespace field is not present in the *config* file (refer the image), then it'll take the namespace called "default" by default.

- But, whatever namespace you give (override) like the below

  kubectl get pod --namespace=<namespace name other than context default one>

  the **user** and **cluster** will be same only.

- **NOTE** **namespace** field will not be there inside the ***config*** file. -----
- You can create as many context you want.
  - Same cluster + different namespaces : new context

    (**optional**, if you want to maintain different contexts; otherwise you can just override the default namespace and move on in the same context using

    --namespace=<other namespace name>
  - Same cluster + different user : new context

    (its **mandatory**)

    you cannot switch user independently, only you can switch *context*.
  - Different cluster : new context

    (its **mandatory**)
- **kubectl config --help**

```
Available Commands:
  current-context   Display the current-context
  delete-cluster    Delete the specified cluster from the kubeconfig
  delete-context    Delete the specified context from the kubeconfig
  delete-user       Delete the specified user from the kubeconfig
  get-clusters      Display clusters defined in the kubeconfig
  get-contexts      Describe one or many contexts
  get-users         Display users defined in the kubeconfig
  rename-context    Rename a context from the kubeconfig file
  set               Set an individual value in a kubeconfig file
  set-cluster       Set a cluster entry in kubeconfig
  set-context       Set a context entry in kubeconfig
  set-credentials   Set a user entry in kubeconfig
  unset             Unset an individual value in a kubeconfig file
  use-context       Set the current-context in a kubeconfig file
  view              Display merged kubeconfig settings or a specified kubeconfig file
```

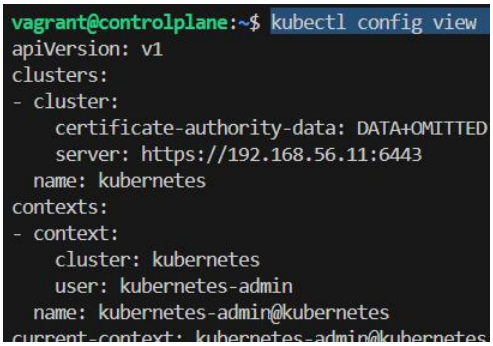
  - **use-context** is used to set a new **current-context** (see in yaml).
- If you want to create a new context

  kubectl config set-context test-context --cluster=kubernetes --user=kubernetes-admin --namespace=test

```
vagrant@controlplane:~$ kubectl config set-context test-context --cluster=kubernetes --user=kubernetes-admin --namespace=test
Context "test-context" created.
vagrant@controlplane:~$ kubectl config get-contexts
CURRENT   NAME                          CLUSTER      AUTHINFO           NAMESPACE
*         kubernetes-admin@kubernetes   kubernetes   kubernetes-admin
          test-context                  kubernetes   kubernetes-admin   test
```

々 You can use this command to view all the configs.

kubectl config view



[cropped, it'll display the whole config file]

々 If you want to change the ***current-context*** then

kubectl config use-context <context name**>**

々 ---------- THE END OF *CONFIG* ----------

々 There is another type of resource called ResourceQuota which is used to set the limit of a specific namespace.

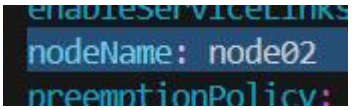
```yaml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: compute-quota
  namespace: dev
spec:
  hard:
    pods: "10"
    requests.cpu: "4"
    requests.memory: 5Gi
    limits.cpu: "10"
    limits.memory: 10Gi
```

々

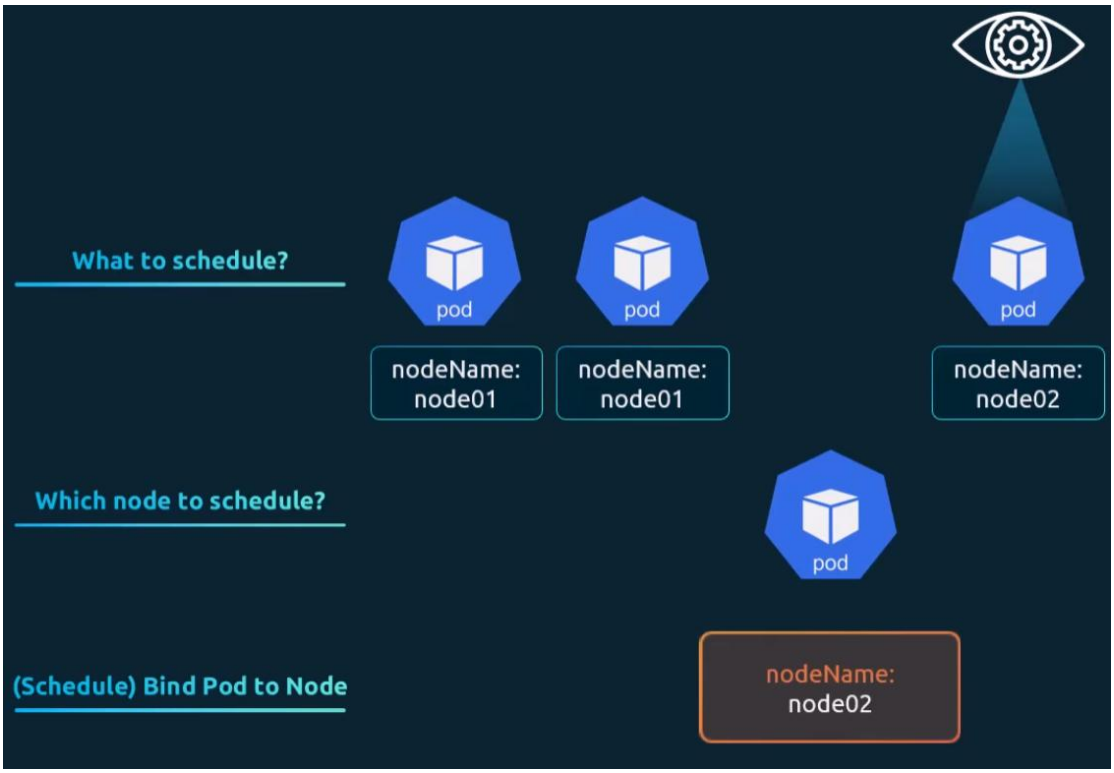
Scheduling

- In a running pod, when you see it, there will be a field called **nodeName** will be present, which contains the worker node's name where that specific pod is running.

  Just execute *kubectl edit pod* <pod name> to see this.



- How ***scheduling*** works?



  - When you run a pod (without specifying *nodeName* field), the pod object is created.
  - Then, *scheduler* sees this, and assign a node to it (***nodeName***).

    NOTE: this ***nodeName*** field is under ***spec*** field of *Pod's* config.
  - Then, one **binding** will be created *in the target node* which will bind the *pod* to the *target node*.

- If the Scheduler doesn't exist,
  - then *nodeName* will not be appended to the pod's config.

    Even if you give *nodeName*, ***binding*** object will not be created in the

target node unless manually send the POST request to the target worker node to create Binding object.

- In this case, the Pod will be in pending state.

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: test-pod
  name: test-pod
spec:
  containers:
  - image: nginx
    name: test-pod
  nodeName: node03
```

```
Every 2.0s: kubectl get pod

NAME                      READY   STATUS    RESTARTS   AGE
ng-dep-66b9cb6d4d-4r822   1/1     Running   0          38h
ng-dep-66b9cb6d4d-f4gl4   1/1     Running   0          38h
ng-dep-66b9cb6d4d-f52lt   1/1     Running   0          39h
test-pod                  0/1     Pending   0          13s
```

  - I simulated is giving a *nodeName* which doesn't exist inside */etc/hosts* file.

    So, the *Binding* object will not be created.

- After sometime, the pod will be **deleted** because its status couldn't be 'Running'.

- Simulating **Manual Scheduling** of a Pod to a Node
  - First create a yaml config of the node giving ***nodeName*** field.

```yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: test-pod
  name: test-pod
spec:
  containers:
  - image: nginx
    name: test-pod
  nodeName: node01
```

  - Now, create the pod

```
vagrant@controlplane:~$ kubectl create -f test-pod.yaml
pod/test-pod created
```

- Now, you need to create a *Binding* yaml file.

```yaml
apiVersion: v1
kind: Binding
metadata:
  name: test-binding
target:
  apiVersion: v1
  kind: Node
  name: node01
```

NOTE: It doesn't contain any data about the Pod, it only contains the details of the target node.

- The yaml is just to understand the key value pairs, in real there is **no such long-lived object called *Binding* exists**.

Its just the JSON payload to be sent in the POST request to bind the pod to the target node.

- Now, you need to trigger the POST request to the **API Server** to bind the pod to the *target worker node*.

NOTE: the worker node doesn't expose any end-points, only API Server in the master node does this.

And, you need to trigger the request to this API Server's end point.

- The POST request looks like this (just an example)

```
curl --header "Content-Type: application/json" \
     --request POST \
     --data '{"apiVersion":"v1", "kind": "Binding", "metadata": {"n
     http://$SERVER/api/v1/namespaces/default/pods/$PODNAME/binding
```

  - You'll get the $SERVER from ***kubectl config view*** command. It'll be in the form

    <API server node's IP>:<API Server's port>

  -

```
vagrant@controlplane:~$ kubectl config view
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: DATA+OMITTED
    server: https://192.168.56.11:6443
```

- $PODNAME is the Pod's name that you get from ***kubectl get pod*** command.

```
vagrant@controlplane:~$ kubectl get pod
NAME                     READY   STATU
ng-dep-66b9cb6d4d-4r822  1/1     Runni
ng-dep-66b9cb6d4d-f4gl4  1/1     Runni
ng-dep-66b9cb6d4d-f52lt  1/1     Runni
test-pod                 1/1     Runni
```

- Now, trigger the request

```
curl --header "Content-Type: application/json" \
 --request POST \
 --data '{"apiVersion":"v1","kind":"Binding","metadata":{"name":"test-binding"},"target":{"apiVersion":"v1","kind":"Node","name":"node01"}}' \
https://192.168.56.11:6443/api/v1/namespaces/default/pods/test-pod/binding/
```

  - It might not work, because you need to provide token or certificates keys in the URL. WILL SEE LATER.

Labels and Selectors

- **labels** and **selectors** are used for grouping and filtering.
- You can give multiple properties inside ***labels*** which will be helpful in filtering the resources properly.
  - You can give as many key value pairs inside *labels* field.
  - 

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ng-dep
  labels:
    name: ng-deployment
    key1: val1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
        key2: val2
    spec:
      containers:
```

    (here I have given 2 labels to each deployment and pod template)
  - Then you can select the resources with

    --selector key1=val1,key2=val2 (you can give any number of label here)

```
vagrant@controlplane:~$ kubectl get all --selector app=nginx,key2=val2
NAME                          READY   STATUS    RESTARTS   AGE
pod/ng-dep-797565664d-dpkjk   1/1     Running   0          80s
pod/ng-dep-797565664d-qljts   1/1     Running   0          80s
pod/ng-dep-797565664d-zgq4k   1/1     Running   0          80s

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/ng-dep-797565664d   3         3         3       80s
```

    NOTE: ReplicaSet created by Deployment will have same label as Pods.
- In case of Deployment or ReplicaSet, whatever you write inside ***matchLabels*** field will be used to get the Pods.
  - All the fields inside *matchLabels* should be there inside the Pod's labels.

- Pod's labels can contain either *same* or *extra* fields than *matchLabels*, but not less.

  If pod doesn't contain all the fields present in *matchLabels* then it'll not be selected.

- Annotations
  - You can also give a field **annotations** inside *metadata* field.
  - Annotations may include *build version, contact information, or other integration data*.
  - ```
    apiVersion: apps/v1
    kind: ReplicaSet
    metadata:
      name: simple-webapp
      labels:
        app: App1
        function: Front-end
      annotations:
        buildversion: "1.34"
    ```
  - Annotations are ideal for *adding tool details, version information, or contact data* without impacting how the system groups or selects objects
    ```

Taints and Tolerations

- Real-life analogy (Assumptions HIT kills cockroaches, but cannot kill Lizards):
  - You want to prevent cockroaches on your bed. So you sprayed HIT around it.
  - Now, Cockroaches cannot come because they are ***intolerant*** to HIT.
  - But, Lizards might come because HIT cannot kill them.
    Or, in other words, lizards are ***tolerant*** to HIT.
- So, How these concepts being used?
  - You put ***taint*** (HIT in the analogy) on the ***nodes***, to prevent the ***pods*** being scheduled to it.
  - Only the ***tolerant*** pods (lizard in the analogy) can be scheduled to the *tainted node*.
  - *Nodes* are ***tainted*** to prevent the *Pods* to be scheduled to them.
    Pods are being ***tolerated*** (when required) to tolerate the ***tainted nodes***.
  - In simple words,
    Taints on nodes **repel pods** that do not have **matching tolerations.**
    Tolerations in pods allow them to **overcome the taint** and be **scheduled on that node.**
- **Taint the Node**

  kubectl taint nodes <node name> **key=value:**<taint-effect>

  [always try to use plural forms i.e. **nodes** ~~not **node**~~]

  Ex: kubectl *taint* nodes node01 env=prod:NoSchedule
  - to remove the taint of the node, just put a **-** at the end.
    kubectl taint nodes <node name> **key=value:**<taint-effect>**-**
    Ex: kubectl *taint* nodes node01 env=prod:NoSchedule**-**
  - Example:
    - Initially no *taint* was there.

      ```
      vagrant@controlplane:~$ kubectl describe node node01 | grep -i taint
      Taints:             <none>
      ```

    - taint node01

      ```
      vagrant@controlplane:~$ kubectl taint nodes node01 env=prod:NoSchedule
      node/node01 tainted
      ```

- node01 got tainted.

```
vagrant@controlplane:~$ kubectl describe node node01 | grep -i taint
Taints:             env=prod:NoSchedule
```

- Remove the taint

```
vagrant@controlplane:~$ kubectl taint nodes node01 env=prod:NoSchedule-
node/node01 untainted
```

- Taint got removed from node01

```
vagrant@controlplane:~$ kubectl describe node node01 | grep -i taint
Taints:             <none>
```

- kubectl taint nodes <node name> key=value:<taint-effect>-

  it is to remove the taint when the exact match will be there.

  - You can also remove the taints for *partial change* by giving this

    kubectl taint nodes <node name> *key-*

    it'll remove the taint of the node where the key is matched.

- When there are multiple taints, it'll be seen like this

```
Taints:             app=product:NoSchedule
                    env=dev:NoSchedule
```

- Tolerate the Pod

  - To tolerate a pod, there is no *imperative* command, also you **cannot** update using ***kubectl edit pod*** <pod name> command.

    you need to configure the yaml.

  - Inside the Pod's ***spec*** field, you can write ***tolerations***

```
spec:
  tolerations:
  - key: "env"
    operator: "Equal"
    value: "prod"
    effect: "NoSchedule"
```

[**double-quote** is mandatory]

    - Operator: "Equal"

      because while giving taint, we wrote ***env=prod***

- You can see, when you create a cluster and add master and worker nodes, pods will not be scheduled to master node.

  It is because master node has a *taint*

```
Taints:             node-role.kubernetes.io/control-plane:NoSchedule
```

  and pods don't have *tolation* to this *taint*.

- **Taint Effects** ----------

- **NoSchedule** (we used this previously)
  - New Pods will not be scheduled on this node *if they are* ***not having the toleration***.
  - Existing Pods will remain as it is. (***even if they doesn't have the toleration***)
  - In simple terms:

    New Pods: ❌

    Existing Pods: ✅
  - In our analogy, you put HIT around of the bed. Cockroaches who are already on the bed will survive but no outside cockroackes can come.
- **PreferNoSchedule**
  - Kubernetes will try to avoid this node to schedule the pods.
  - But, it might schedule if there is no other option.
  - In simple terms:

    New Pods: ❌✅

    Existing Pods: ✅
- **NoExecute**
  - New Pods will not be scheduled to this node (same as *NoSchedule* effect)
  - Existing Pods will be removed (different from *NoSchedule* effect)
  - In simple terms:

    New Pods: ❌

    Existing Pods: ❌
  - In our analogy, you put HIT around the bed and also on the bed. Now cockroaches will be dead and no outside cockroaches can come. Lizards are safe.
- NOTE: All the not-scheduling and delete from the node things which are written above, those are for the pods who don't have *tolerance* to that *taint*. If the pods are having *tolerance*, then they (new and existing) will not be affected.
  - ----------

    --

nodeSelector and nodeAffinity

- While *taint* and *tolerations* were to avoid a node to be scheduled for pods, ***nodeSelector*** and ***nodeAffinity*** are there to schedule a node for the pods.
- nodeSelector ----------

- It works in the same principle ***selector***.
- First provide the ***labels*** to the node. [before we labelling the Pods]

```
kubectl label nodes node1 env=prod
```

- Use ***nodeSelector*** field to match the Node labels. [ReplicaSet's *selectors* field]

```
spec:
  nodeSelector:
    env: prod
```

- Example
  - Created a 2 pod yaml files (one for env=prod, another for env=dev)

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: prod-pod
  name: prod-pod
spec:
  nodeSelector:
    env: prod
  containers:
  - image: nginx
    name: prod-cont
    ports:
    - containerPort: 80
      name: nginx-port
```

```
apiVersion: v1
kind: Pod
metadata:
  labels:
    app: dev-pod
  name: dev-pod
spec:
  nodeSelector:
    env: dev
  containers:
  - image: nginx
    name: dev-cont
    ports:
    - containerPort: 80
      name: nginx-port
```

  - Label ***node01*** as ***env=dev***, and ***node02*** as ***env=prod***

```
vagrant@controlplane:~$ kubectl label nodes node01 env=dev
node/node01 labeled
vagrant@controlplane:~$ kubectl label nodes node02 env=prod
node/node02 labeled
```

```
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    env=dev
```
[node01]

```
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    env=prod
```
[node02]

- Created prod pod and it was scheduled in node02

```
vagrant@controlplane:~$ kubectl create -f prod-pod.yaml
pod/prod-pod created
vagrant@controlplane:~$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE   IP              NODE     NOMINATED NODE   READINESS GATES
prod-pod   1/1     Running   0          8s    10.244.140.103  node02   <none>           <none>
```

- Created dev pod and it was scheduled in node01

```
vagrant@controlplane:~$ kubectl get pods -o wide
NAME       READY   STATUS    RESTARTS   AGE   IP              NODE     NOMINATED NODE   READINESS GATES
dev-pod    1/1     Running   0          36s   10.244.196.162  node01   <none>           <none>
```

nodeAffinity ----------

- The *nodeSelector* was very straight forward. You need to give *label* properly

  [hard selection just like *matchLabels* in ReplicaSet]

  nodeAffinity is having multiple types of label mathing. [like *matchExpressions* in ReplicaSet]

  - What if you want to say like

    place the pod *where env=prod or env=test*

    place the pod *where env != dev*

    in this case, ***nodeAffinity*** can be used.

- The yaml looks like the below

```yaml
spec:
  containers:
    - name: data-processor
      image: data-processor
  affinity:
    nodeAffinity:
      requiredDuringSchedulingIgnoredDuringExecution:
        nodeSelectorTerms:
          - matchExpressions:
              - key: size
                operator: In
                values:
                  - Large
```

```yaml
preferredDuringSchedulingIgnoredDuringExecution:
- weight: 5
  preference:
    matchExpressions:
    - key: zone
      operator: In
      values: [us-east-1]
```

- 2 types of nodeAffinity

  requiredDuringSchedulingIgnoredDuringExecution

  preferredDuringSchedulingIgnoredDuringExecution

- Pods are having 2 stage: ***scheduling, execution***
  - When the pod is not created in any node, it is first scheduled. [**Scheduling**]
  - After the pod being scheduled, it'll be created in the scheduled node. [**Execution**]
  - Currently, nodeAffinity is only there for *scheduling* phase, you can see its written ***Ignored***during***Execution***.
- *required* one contains ***nodeSelectorTerms*** whereas *preferred* one contains ***weight*** and ***preference*** .
- In case of *requiredDuringSchedulingIgnoredDuringExecution*, if the expression doesn't match with any node, then the Pod will not be scheduled at all. [**HARD** matching]

In case of *preferredDuringSchedulingIgnoredDuringExecution*, all the nodes will be checked and weight will be calculated. Highest preferred node will be scheduled for the pod. [**SOFT** matching]

- How the *weight* is calculated in case of *preferredDuringSchedulingIgnoredDuringExecution* ?
  - If you have provided some preferences, all will be checked in all the nodes available.

    Each preference will be having some weight (you'll define this)
  - The nodes will be checked for all the preferences. If preference matches in that node then weight for that node will be increased with the weight provided for that preference.
  -

```
preferredDuringSchedulingIgnoredDuringExecution:
- weight: 5
  preference:
    matchExpressions:
    - key: env
      operator: In
      values: [prod]

- weight: 3
  preference:
    matchExpressions:
    - key: disk
      operator: In
      values: [ssd]
```

  -

Node	env=prod	disk=ssd	Score
node1	✅	✅	5 + 3 = 8
node2	✅	❌	5 + 0 = 5
node3	❌	✅	0 + 3 = 3

    here, *node1* has both the preference matched so its weight wil be 8, so pods will be scheduled to *node1*.

- Use **kubectl explain pod.spec.affinity.nodeAffinity --recursive** to get the fields and directly paste inside pod yaml file.

- Here we used ***matchExpressions*** to match the label of the nodes, there is another called ***matchFields***.
  - It matches any type of fields you want, not just *labels*.

```
matchFields:
- key: metadata.name
  operator: In
  values: [node1]
```
[metadata.name must be node1]

- There are other 2 types of affinity,

  podAffinity, **podAntiAffinity**
  - *nodeAffinity* checks the labels in the nodes and match that.
  - *podAffinity* and *podAntiAffinity* checks the lebels of the pods which are running inside the node.
- ***podAffinity***
  - The pod running on target nodes will be checked with the preferences/nodeSelectorTerms.

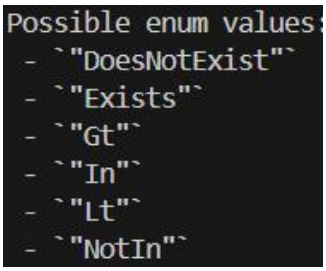
    if matches then that node will be scheduled.
- ***podAntiAffinity***
  - Opposite of *podAffinity*,

    the lowest matched (in case of *preferredDuring*....) will be selected.

    or not matched (in case of *requiredDuring*...) will be selected.
- Some operators of *matchExpressions*
  -
```
- matchExpressions:
    - key: size
      operator: In
      values:
        - Large
```
[In]
  -
```
- matchExpressions:
    - key: size
      operator: Exists
```
[Exists]
  - All the operators are

    kubectl explain pod.spec.affinity.nodeAffinity.preferredDuringSchedulingIgnoredDurin

gExecution.preference.matchExpressions



NodeAffinity configurations ----------

- *spec.affinity.nodeAffinity* has to be updated.

  nodeAffinity support complex queries like DoesNotExist, Exists, Gt, In, Lt, NotIn

- nodeAffinity is of 2 types:
  - *requiredDuringSchedulingIgnoredDuringExecution* [HARD rule]
  - *preferredDuringSchedulingIgnoredDuringExecution* [Scored basis]

```
nodeAffinity
├── preferredDuringSchedulingIgnoredDuringExecution []
│   ├── preference
│   │   ├── matchExpressions []
│   │   └── matchFields []
│   └── weight
└── requiredDuringSchedulingIgnoredDuringExecution
    └── nodeSelectorTerms []
        ├── matchExpressions []
        └── matchFields []
```

-
  - [~~its not *match**Labels***~~, its *match**Fields***]
  - *matchExpressions* is to query labels of nodes.
  - *matchFields* is to query any fields. [Ex: *metadata.name*]
- *matchFields* and *matchExpressions* [both are *list*]

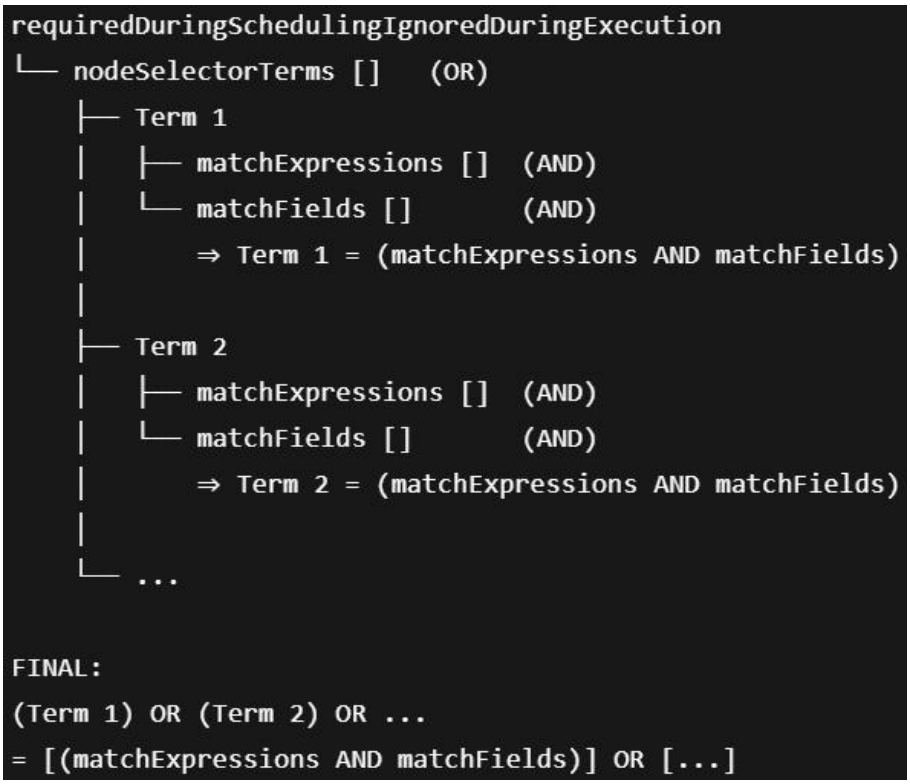
```yaml
matchExpressions:
- key: env
  operator: In
  values:
  - prod
- key: disk
  operator: In
  values:
  - ssd
```

```yaml
matchFields:
- key: metadata.name
  operator: In
  values:
  - node01
  - node02
- key: metadata.resourceVersion
  operator: In
  values:
  - "353007"
  - "353008"
```

-
  - *matchExpressions*: query1 **AND** query2 **AND** ...

    matchFields: query1 **AND** query2 **AND** ...

- *requiredDuringSchedulingIgnoredDuringExecution*



  - nodeSelectorTerms: *term1* ***OR*** *term2* ***OR*** ...
  - Each term: *matchFields* ***AND*** *matchExpressions*

    matchFields: *query1* ***AND*** *query2* ***AND*** ...

    matchExpressions: *query1* ***AND*** *query2* ***AND*** ...
  - Any one term should match.

    Means, all the *matchExpressions* and *matchFields* query of that terms should match.
  - *nodeSelectorTerms = [{matchExpressions AND matchFields} OR {..} OR ..]*
  - If you mark here, *nodeSelectorTerms* is not even required. *requiredDuri...Execution* could have made a list instead of a dictionary.

- *preferredDuringSchedulingIgnoredDuringExecution*

```
preferredDuringSchedulingIgnoredDuringExecution []   (scored list)
├── Item 1
│   ├── preference
│   │   ├── matchExpressions []  (AND)
│   │   └── matchFields []       (AND)
│   │       ⇒ match = (matchExpressions AND matchFields)
│   └── weight
│
├── Item 2
│   ├── preference
│   │   ├── matchExpressions []  (AND)
│   │   └── matchFields []       (AND)
│   │       ⇒ match = (matchExpressions AND matchFields)
│   └── weight
│
└── ...

FINAL (scoring, not filtering):
score(node) = sum(weight of each item that matches)
```

  - Here, preferredDu....Execution is list. No unnecessary extra child field like the previous.
  - [

    {preference: {matchExpressions AND matchFields}. weight: calculate},

    {...},

    ...

    ]
  - For a item:
  - weight=provided value (if matchExpressions AND matchFields) or 0 (zero)
  - Total weight: item1 weight + item2 weight ...
  - the node who gets more weight will be selected for the pod.
- requiredDur....Execution: at least one should match

  preferredDur...Execution: more matches, more score
- ---

Resource Requirements

- There are 2 things, **Request** and **Limit**.
  - Request: minimum required resource (used for scheduling)
  - Limit: maximum allowed resource (enforced at runtime)
- The YAML look like this.

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: demo
spec:
  containers:
  - name: app
    image: nginx
    resources:
      requests:
        cpu: "500m"
        memory: "256Mi"
      limits:
        cpu: "1"
        memory: "512Mi"
```

[resources is written inside *containers*, not spec] < ----------

- **requests** (Scheduling Decision)
  - If any node has at least this much free resources, then only that node can be scheduled for the pod.

    Otherwise it'll not be scheduled.
- **limits** (Runtime enforcement)
  - CPU: throttled (if exceed than the limit set) [throttled means slow down]
  - Memory: container is ***killed*** (OOMKilled) (if exceed than the limit set)
- CPU vs Memory behaviour
  - CPU can exceed request, but Memory cannot.

- In case of hitting the limit,

  CPU: throttled [slow down]

  Memory: *container killed*

- Resource is not there, then by default it takes as 0.

  But, if resource is not there and limit is there: resource will be taken same as limit.

  If limit is not there, then it'll assume the pod can use ***unlimited*** resources.

- **1** (one) count of CPU is same as
  - 1 AWS **v**CPU
  - 1 GCP Core
  - 1 Azure Core
  - 1 Hyperthread
- Memory
  - **1 G** : 1 Gigabyte
  - **1 M** : 1 Megabyte
  - **1 K** : 1 Kilobyte
  - **1 Gi** : 1 Gibibyte
  - **1 Mi** : 1 Mebibyte
  - **1 Ki** : 1 Kibibyte

    [**K** : 1000 bytes, **Ki**: 1024 bytes ; K is binary (2's power), Ki is decimal (10's power)]

- **cpu** lowest unit: **1m** (1 mili CPU or 0.001 CPU) [1 meanns one full CPU]

  memory lowest unit: **1** (1 byte) [only use Gi, Mi, Ki; ~~don't use G, M, K~~]

- We can have 4 scenarios:

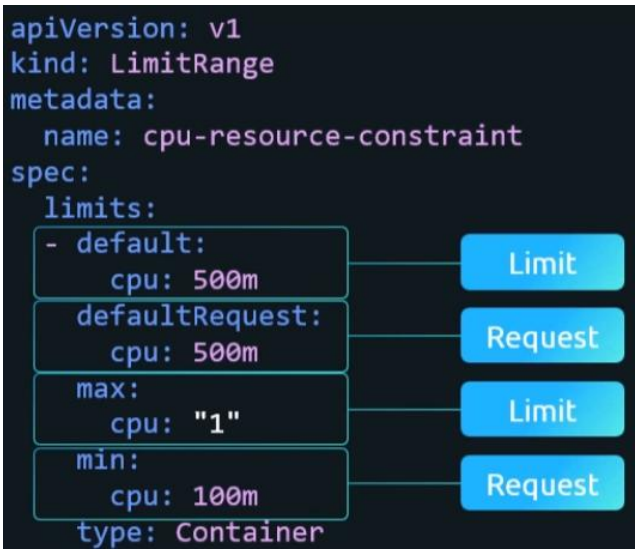
  No requests, No limits

  No requests, Limits

  Requests, Limits

  Requests, No limits
  - In the first case, if one pod (container inside pod because requests and limits are set per container, not per pod) is consuming all the CPU, and another pod is also assigned to same node (because if request is not given, it'll be taken as 0) which is not able to get the resources.
  - In the second case, request will be taken as same as **limits**.

- In the 3rd case, the pods are having both request and limits. Lets say in this case pod-1 doesn't need extra resources but pod-2 needs.
  And we have set the limit for both the pods which will stop *pod-2* to use more resources. Which is not ideal
- In the 4th case, the problem of *3rd* case can be solved. Because if a pod is being assigned to a node means the node has required amount of resources available that the pod needs.
  And, there is no limit so pod can use as much as it want.

- There is something called **LimitRange**, it by default injects the ***requests*** and ***limits*** to the pods if not specified in the yaml.
  - It works in **namespace** level.
  - 

  - Multiple LimitRange can be configured in a namespace, but they should not conflict.
    Ex: one says max cpu: 500m, another says min cpu: 600m
    in this case, pod will not be created because of conflicting values.
    In real life: **one LimitRange per namespace is maintained**.
  - **default** :
    - it is the ***limits*** field.
    - If the pod yaml doesn't contain ***limits*** fields, then this LimitRange's *spec.limits.default* will be added there.
  - **defaultRequest** :
    - It is the ***requests*** field.

- If the pod yaml doesn't contain ***requests*** fields, then this LimitRange's *spec.requests* default will be added there.

- **min**
  - The ***requests*** or ***limits*** shuld not be less than this..
  - If the container (inside pod) asks lesser resource then this *min* then it'll be rejected.
- **max** :
  - The ***requests*** or ***limits*** shuld not be more than this.
  - The container cannot request more requests then that defined in *max*.
- **type** : Container

- **DaemonSet**
  - It is similar to ***ReplicaSet***
  - It makes sure that each node should contain one copy of the pod specified in *DaemonSet* configuration.
  - Earlier it used to append ***nodeName*** field and POST request to create the container in the target node.

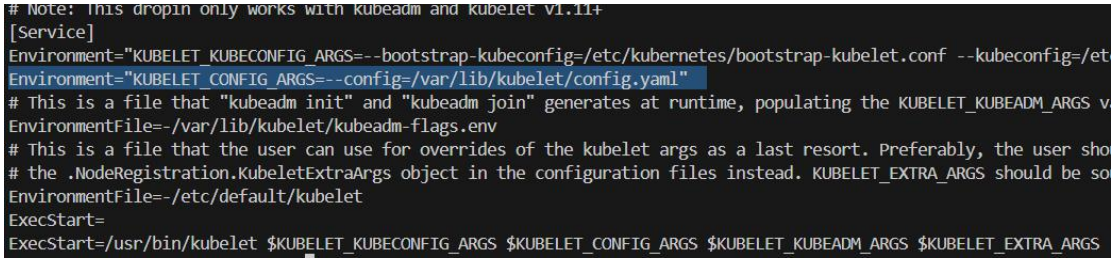
    But now, it uses ***nodeAffinity*** to assign a specific node to the Pod.
  - The yaml is exact same as that of *ReplicaSet*, just the ***kind*** will be different.

```yaml
apiVersion: apps/v1
kind: DaemonSet
metadata:
  name: monitoring-daemon
spec:
  selector:
    matchLabels:
      app: monitoring-agent
  template:
    metadata:
      labels:
        app: monitoring-agent
    spec:
      containers:
      - name: monitoring-agent
        image: monitoring-agent
```

  -
- F

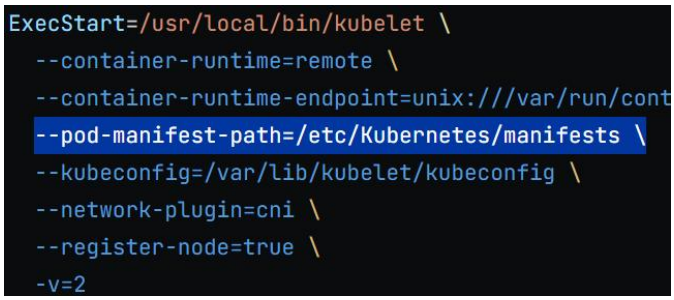
Static Pods

- Pods are created by the command of *kube-api-server*, which is listened by *kubelet* (in worker nodes) and create and start the container.
  what if the worker node is not connected to any Kubernetes cluster, it just has ***kubelet*** installed. How to run a pod here?
- You need to find the ***staticPodPath*** in the node where ***kubelet*** is present. ***kubelet*** constantly see this path and create and update the pod if any ***yaml*** config is present inside this path.
- **NOTE :-** You can only create pods in this way, other resources like *ReplicaSet, Deployment* cannot be created because they are completely of *Kubernetes* control plane things.
- To find the ***staticPodPath*** the below methods are there [all in worker node, not master node]
  - **Method-1**
    - *systemctl cat kubelet*



(you'll get something like this; this image is cropped)

if you find something like below, then see the **--pod-manifest-path**



- Open that ***config.yaml*** file present there, you'll find a key called *staticPodPath*

- In this path i.e. ***/etc/kubernetes/manifests***, if you create a Pod's yaml file, then ***kubelet*** will create that pod inside that node.
- The above displayed is ***kubelet.service*** file (it is same as the above *systemctl cat kubelet*)
- Its mostly in */usr/lib/systemd/system/kubelet.service*
  if not, then find other service file directories like /etc/systemd/system, */var/lib/systemd/system* ..etc
- There you'll find the ***ExecStart*** field

```
[Unit]
Description=kubelet: The Kubernetes Node Agent
Documentation=https://kubernetes.io/docs/
Wants=network-online.target
After=network-online.target

[Service]
ExecStart=/usr/bin/kubelet
Restart=always
StartLimitInterval=0
RestartSec=10

[Install]
WantedBy=multi-user.target
```

- **Method-2**
  - **ps -ef | grep kubelet**

```
vagrant@node01:/usr/local/bin$ ps -ef | grep kubelet
root        3762       1  2 May02 ?        00:32:17 /usr/bin/kubelet --bootstrap-kubeconfig=/etc/kubernet
es/bootstrap-kubelet.conf --kubeconfig=/etc/kubernetes/kubelet.conf --config=/var/lib/kubelet/config.yaml
 --node-ip 192.168.56.21
```

  - You'll find the ***config.yaml*** path here.
  - Go to that *yaml* file and get the ***staticPodPath***.

- **To check in the worker node**
  - You can use ***crictl*** command to see if the container is running.
    You can't use ***kubectl*** command there because its not kubernetes.
  - *crictl* is used to talk to *containerd*.
    If you get "Permission Denied" then run with *sudo*.
  - **sudo crictl ps**

```
vagrant@node01:~$ sudo crictl ps
CONTAINER           IMAGE               CREATED             STATE               NAME                ATTEMPT             POD ID              POD                 NA
306e8944e30c3       6c3a6ea6608c8       26 minutes ago      Running             static-cont         0                   37b86e808bd54       test-static-pod-node01  de
```

  - kubeadm creates ***static pods*** in master node to run the *control plane resources*.

```
vagrant@controlplane:/etc/kubernetes/manifests$ ls
etcd.yaml  kube-apiserver.yaml  kube-controller-manager.yaml  kube-scheduler.yaml
```

- When you execute ***kubectl get pods*** in the control plane, you'll see the *static pod* there.

  HOW? Because this should be specific to the worker node and kube API Server shouldn't know about this. Read about the ***end-to-end flow*** (next page) for this.

| End-To-End flow from Pod creation command to actual Running of Pod |

---------- IMPORTANT ----------

- Below is the flow end-to-end
  - User creates a Pod
    - *kubectl apply -f my-pod.yaml* (or) *kubectl run …*
    - This request goes to API Server.
  - API Server
    - It validates the request.
    - Store the *Pod* object (PodSpec) in *etcd*.
  - Scheduler
    - It sees the unscheduled pod in *etcd*.
    - It schedule any node to this pod [updates ***spec.nodeName***]
  - Kubelet (in the scheduled node)
    - Watches API Server, and see a pod is assigned to it.
      [kubelet opens a *watch* connection. API Server streams updates through it]
    - Pulls the ***PodSpec***.
  - Kubelet → Container runtime (*containerd*)
    - Creates Pod sandbox (pause container: otherwise container will get its own IP; instead of using Pod's IP)
    - Creates all the required containers defined in *PodSpec*.
  - Container Runtime (containerd)
    - Pulls images, & creates the containers.
    - Assign *container IDs*.
    - Start the processes.
  - Kubelet (after creation)
    - Maps *Pod ID → Container IDs*
    - Monitors the containers
      running / crashed / restarted
  - Kubelet → API Server (status update)
    - API server doesn't poll for Pod's status.
      Kubelet update API server about the Pod's status.
    - API Server stores updated status in ***etcd***.

```
status:
  phase: Running
  containerStatuses:
    - name: nginx
      containerID: containerd://abc123
      state: running
```

- Control Plane (now updated)
  - ***etcd*** contains

    PodSpec (desired state)

    PodStatus (actual state from Kubelet)
- Controllers (ReplicaSets, etc..)
  - Watch API Server
  - Compares

    Desired state vs Current state
  - Take action if mismatch.

- Now, the question arises: Does the ReplicaSet or other controllers repeatedly poll API Server about the pod's status?
  - No, it starts a ***watch event***.
  - Its not a *web-hook*, it's a long-lived connection (***streaming connection***)
  - Whenever something changes, API Server streams it, so *controllers* get to know about the pod's status.
- *Control Plane* knows nothing about the containers.

  It gets the details about the containers (id and status) via Kubectl only.

Component	Knows PodSpec	Knows Container ID	Runs Containers
API Server	✅	⚠️ (reported only)	❌
etcd	✅	⚠️ (stored only)	❌
kubelet	✅	✅ (authoritative)	❌
containerd	❌	✅ (real)	✅

- To see the details about the containers in *Control Plane* and *Worker Node*
  - When you execute

    kubectl get pod <pod name> **-o yaml**

    (or)

    kubectl edit pod <pod name>
    - It shows the content of ***etcd*** in yaml format.

- **status.containerStatuses**

  this field contains the list of containers associated with that pod along with their IDs, image, state (running, etc.), volumes ..ec.

```
containerStatuses:
- containerID: containerd://98243133419e0d6
  image: docker.io/library/nginx:1.29
  imageID: docker.io/library/nginx@sha256:6
  lastState: {}
  name: dev-cont
  ready: true
  resources: {}
  restartCount: 0
  started: true
  state:
    running:
      startedAt: "2026-05-02T18:10:45Z"
  user:
    linux:
      gid: 0
      supplementalGroups:
      - 0
      uid: 0
  volumeMounts:
  - mountPath: /var/run/secrets/kubernetes.
```

NOTE ----------

- ***kubelet*** has a watch connection with API Server.

  And, whenever API Server gets some request (either create a new pod or update an existing pod), it updates the ***etcd*** .

- When you trigger *kubectl edit pod <pod name>*
  - API Server gets the update request, it updates the *etcd* if there is any changes.
  - *kubelet* already has a *watch connection*, through that API Server sends the modified *PodSpec*.
  - *kubelet* gets the new PodSpec and compare with the existing one.
  - If any changes, then it recreates the container (if allowed; if not allowed then reject it)

    it sends the report of new container to the API Server i.e. new ***Container ID***, ..etc.

```
containerStatuses:
- containerID: containerd://3d835f86cfac50fe61ea18455f9b7a9a22f98e73aa22d2c34dd54d4c7bf9b76b
  image: docker.io/library/nginx:1.29-perl
  imageID: docker.io/library/nginx@sha256:3d5526f7dc2dfd0b6cbecbc664ca2d7ab1e1b5f572a289c6a7607728c5f5b85c
  lastState:
    terminated:
      containerID: containerd://98243133419e0d62baa1134c86c58fa102f93170f3ff7f82e08ebce518078c96
      exitCode: 0
      finishedAt: "2026-05-03T13:08:32Z"
      reason: Completed
      startedAt: "2026-05-02T18:10:45Z"
```

[you can see, *lastState.terminated*; it contains previous containerID here]

[contains only last terminated container details, not all]

- In case of no change, doesn't do anything.

- In case of ***static pod***, *kubelet* sends the same report to API Server; but with some extra fields
  - The below is the *metadata* of normal pod (pod created by api server)

```
metadata:
  annotations:
    cni.projectcalico.org/containerID: c6d2a3016d3a055cededa48f1ca2c97a5c69303ce534161abd71d26a4e4c2641
    cni.projectcalico.org/podIP: 10.244.196.162/32
    cni.projectcalico.org/podIPs: 10.244.196.162/32
```

  - The below is the *metadata* of ***static pod*** (pod created in worker node itself)

```
metadata:
  annotations:
    cni.projectcalico.org/containerID: 37b86e808bd5439f78bf763fa7914144be4b67de4342957d3253bc2412eaa40a
    cni.projectcalico.org/podIP: 10.244.196.163/32
    cni.projectcalico.org/podIPs: 10.244.196.163/32
    kubernetes.io/config.hash: af6e265c669e39311b5f3774274e0832
    kubernetes.io/config.mirror: af6e265c669e39311b5f3774274e0832
    kubernetes.io/config.seen: "2026-05-03T09:51:34.943426976Z"
    kubernetes.io/config.source: file
```

  - You can see 2 key value pairs:

    kubernetes.io/config.mirror: <hash>

    kubernetes.io/config.source: file

    these tells API Server that this is not a Pod created by API Server; this is a *mirror* of a static pod.
  - *config.source: file* means the pod is created from a *file* present in the *node* itself;

    if this field is not there means the *pod* is created by *API Server* itself.

- The API server stores mirror pod objects also in etcd, but they are not the source of truth. They are copies of static pods created by kubelet and marked with annotations to indicate their origin.

*Resources inside Pods **vs** Resources as normal Processes*

- As we saw the control plane processes run in ***static pods*** in the control plane node when you configure the cluster via ***kubeadm***, how the things communicate with eachother then?

  And, how the resources communicate with each other when they are not running inside any *pod* rather running outside normally as a process?

- **First lets see, how they work when run as normal processes.**
  - There are multiple ***configuration*** files inside ***/etc/kubernetes*** directory.

    ```
    vagrant@controlplane:/etc/kubernetes$ ls
    admin.conf  controller-manager.conf  kubelet.conf  manifests  pki  scheduler.conf  super-admin.conf
    ```

  - All these files contains the specific requirements of each processes like *scheduler, controller manager* ...etc.
  - All the ***.conf*** files are having same structure as the ***~/.kube/config*** file, but

    user, certificates, permissions are different.
  - In the ***scheduler.conf*** the user is system:kube-scheduler

    ```
    users:
    - name: system:kube-scheduler
      user:
        client-certificate-data: ...
    ```

  - *kubelet, kubectl, controller, scheduler* all talks to API Server.

    API Server only directly talks to ***etcd***.

    ```
    scheduler ──► API Server ──► etcd
    controller ─► API Server
    kubelet ────► API Server
    kubectl ────► API Server
    ```

    [No component talks to ***etcd*** directly except *API Server*]
  - For example, if you see the *config* files, the ***server*** will be having

    <control plane node's IP>:6443

    because, API Server listens on port *6443*.

    ```
    vagrant@controlplane:/etc/kubernetes$ sudo cat admin.conf | grep server
        server: https://192.168.56.11:6443
    vagrant@controlplane:/etc/kubernetes$ sudo cat scheduler.conf | grep server
        server: https://192.168.56.11:6443
    vagrant@controlplane:/etc/kubernetes$ sudo cat kubelet.conf | grep server
        server: https://192.168.56.11:6443
    ```

- All the server contains this end-point to talk to [because all talk to *API Server* only]

  And, all the communication happens via *HTTP Calls*.
- This is how the communication and work happens in ***control plane*** nodes.

- **Now, how the resources behaves when all are present inside Pods**
  - As we saw, all the resources communication happens via *HTTP Calls* only.

    So, even if they are present inside the pods, there will some port mapping like

    6443:6443 (just example)
  - So, nothing changes. All the end-points remains same.
  - ***kubeadm*** keeps the same port mapping from host to pods, so the end-points doesn't change.

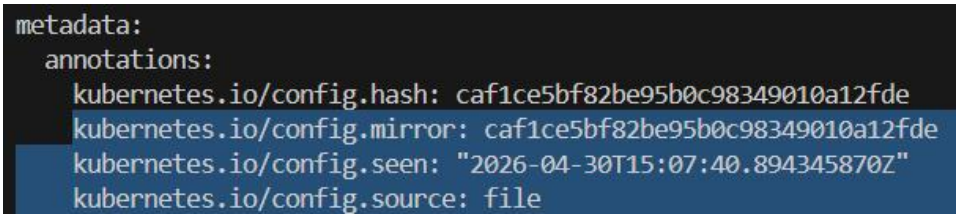
    In case if you want to bind different host port to the pod, then ***configuration*** *files needs to be updated with this*.

    And, all of the things will work now.
  - ***kubeadm*** runs all the resources in ***static pods*** (not normal pods).

    So, *API Server* or *etcd* has nothing to do with this.

    [[[as we know, even if the pods are static, when you execute ***kubectl get pods*** command you'll be able to see the static pods because ***kubelet*** informs *API Server* about the static servers as well and *API Server* stores the details about the *static pods* inside ***etcd***; but the *static pods* are having 2 extra fields inside ***annotations*** which are *config.mirror: <hash>* and *config.source: file* **]]]**

    when you see the details about the *resources pods*, you'll find the fields inside *annotations*.

    (kube-scheduler pod)

- So, API Server cannot update these pods because *static pod* can only be updated if you change the ***yaml*** files directly inside *manifest* (or any directory configured) directory.
- The main advantage of running the resources inside *static pods* is if the pods crashes or the resources fails, kubelet will **automatically restarts** the pods by itself.

- **NOTE ::::** ***kubelet*** doesn't run inside any pod, it is a ***systemd*** service;
  So, when you execute ***systemctl cat kubelet*** you get the *kubelet.service* file content.
  But, it'll not work for kube-apiserver, etcd, kube-scheduler ..etc because they run inside Pods.

Multiple Schedulers

- There are 3 different files, that might be confusing but they are different.
  - **kube-scheduler.yaml**

    present inside */etc/kubernetes/manifest* (Static Pod's yaml) **kind: Pod**
  - **scheduler.conf**

    kube-scheduler config file (same structure as of ~/.kube/config file)
  - **scheduler-config.yaml**

    [not same as *scheduler.conf*] **kind: KubeSchedulerConfiguration**

    it is the yaml file to create ***KubeSchedulerConfiguration*** resource.

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: default-scheduler
```

  - In simple words:

    kube-scheduler.yaml creates the pod to run scheduler within it.

    scheduler.conf is the configuration file (details and certificates) for that scheduler.

    scheduler-config.yaml is the extra configuration plugins etc.

- When you open the ***kube-scheduler.yaml*** (kind: Pod, where kube-scheduler runs), you'll see a ***command*** field (which is run inside the container).
  -

```yaml
spec:
  containers:
  - command:
    - kube-scheduler
    - --authentication-kubeconfig=/etc/kubernetes/scheduler.conf
    - --authorization-kubeconfig=/etc/kubernetes/scheduler.conf
    - --bind-address=127.0.0.1
    - --kubeconfig=/etc/kubernetes/scheduler.conf
    - --leader-elect=true
    image: registry.k8s.io/kube-scheduler:v1.36.0
```

    [**kube-scheduler** commands comes with that *image* mentioned here]
  - Here, the ***scheduler-config.yaml*** is not mentioned, rather directly all the things are written here.

- In modern way, ***scheduler-config.yaml*** is present which is more proper.

```
spec:
  containers:
    - name: kube-scheduler
      image: k8s.gcr.io/kube-scheduler-amd64:v1.11.3
      command:
        - kube-scheduler
        - --address=127.0.0.1
        - --kubeconfig=/etc/kubernetes/scheduler.conf
        - --config=/etc/kubernetes/my-scheduler-config.yaml
```

[the name would be *scheduler-config.yaml* in case of default one]

- After creating your custom scheduler, you need to mention this explicitly inside Pod's yaml file to choose the *custom scheduler* instead of the default one.

```
apiVersion: v1
kind: Pod
metadata:
  name: nginx
spec:
  containers:
    - name: nginx
      image: nginx
  schedulerName: my-custom-scheduler
```

- F

Scheduler Profiles

- Till now we saw the below things:
  - One ***yaml*** file is there to create the Pod containing *scheduler* process (running scheduler binaries) [*/etc/kubernetes/manifest/**kube-scheduler.yaml***]
  - One ***config*** file is there which contains the details about *cluster, user, context* to authenticate and authorize kube-scheduler.
    [*/etc/kubernetes/**scheduler.conf***]
  - One ***yaml*** file is there which contains the *configurations* of the *scheduler*
    [***scheduler-config.yaml***]

```yaml
# my-scheduler-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
```

    [for example]

    it's reference will be there in the *command* section of *kube-scheduler.yaml* (in modern systems) or it'll not be present (in older systems the configs are directly given in that command section)

- The Pod scheduling process comprises of mainly below phases:
  - ***Scheduling queue***

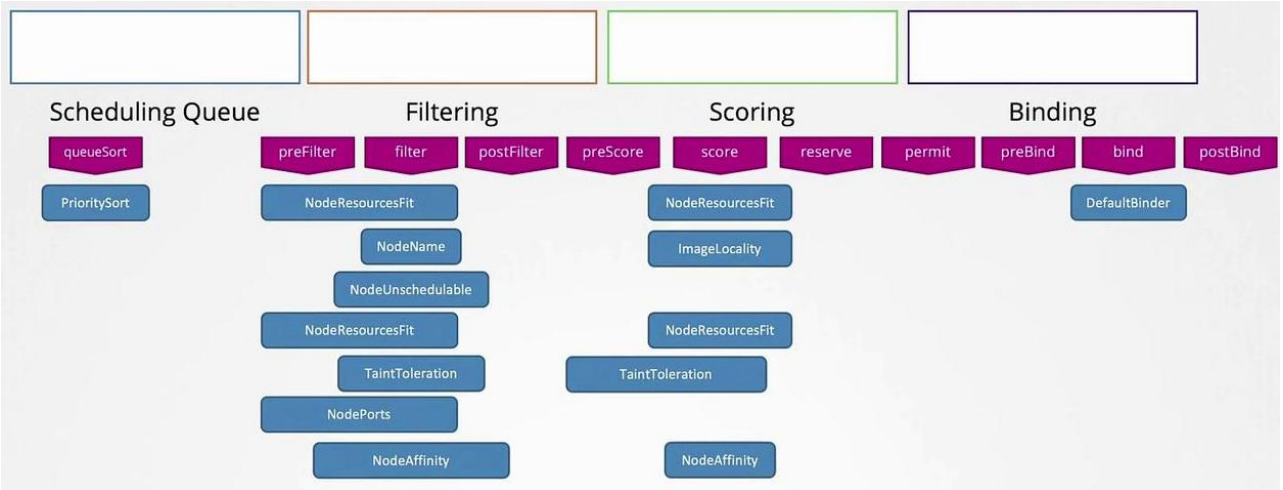
    all the unscheduled pods will be kept in a queue according to their priority.
  - ***Filter phase***

    scheduler eliminates the nodes which do not satisfy the requirement of the Pod's resource requirement.
  - ***Scoring phase***

    nodes that pass *filter phase* are now scored. The node that is scored more will be scheduler.
  - ***Binding phase***

    the node who got scored more score will be binded to the pod.

- There are different ***plugins*** being used during this scheduling process.



  - ***PrioritySort*** *plugin*

    during the scheduling queue phase, this plugin orders the Pods based on their priorities.

  - ***NodeResourceFit*** *plugin*

    it is used during *filter phase* to exclude nodes.

    Also, during *scoring phase* it score the nodes that pass in *filter phase*.

  - ***NodeUnschedulable*** *plugin*

    it makes sure that the nodes marked as *Unschedulable: true* will not get the pod assigned.

```
vagrant@controlplane:/etc/kubernetes$ kubectl describe node node01
Name:               node01
Roles:              <none>
Labels:             beta.kubernetes.io/arch=amd64
                    beta.kubernetes.io/os=linux
                    env=dev
                    kubernetes.io/arch=amd64
                    kubernetes.io/hostname=node01
                    kubernetes.io/os=linux
Annotations:        csi.volume.kubernetes.io/nodeid: {"csi.tigera.
                    node.alpha.kubernetes.io/ttl: 0
                    projectcalico.org/IPv4Address: 192.168.56.21/24
                    projectcalico.org/IPv4VXLANTunnelAddr: 10.244.
                    volumes.kubernetes.io/controller-managed-attach
CreationTimestamp:  Thu, 30 Apr 2026 15:26:20 +0000
Taints:             <none>
Unschedulable:      false
```

  - ***ImageLocality*** *plugin*

    this just gives a *soft preference* if the node already contains the *required container image*.

- ***DefaultBinder*** *plugin*

  in the binding phase, this plugin finalize the Pod-to-Node assignment.

- In the above image, each stages are further divided into sub-stages like

  queueSort

  preFilter, filter, postFilter

  preScore, score, reserve

  permit, preBind, bind, postBind

  - All these are called ***extension points***.
  - With the *extension points*, the ***plugins*** *are binded*.

- We can create multiple ***scheduler pods*** and run as many scheduler as we want.

  And for each scheduler, we can create a ***scheduler configuration*** file (yaml) (kind: KubeSchedulerConfiguration).

  - In this *KubeSchedulerConfiguration*, we can choose which ***plugins*** *we want to* ***enable/disable*** *for the* ***extension points***.
  - Let say you have 2 pods to be scheduled, and you want different types of *plugins* will be used in different *extension points* in those.

    Then, you can create 2 *scheduler pods* and write *2 kube scheduler configuration* each having specific plugins enabled/disabled.

```yaml
# my-scheduler-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
```

```yaml
# my-scheduler-2-config.yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler-2
```

  - But instead of doing this, we can just create one scheduler and attach multiple profiles to it.

    [***scheduler*** is the **process** that is running inside the Pod.

    Profiles are the **configurations having specific modules enabled/disabled** at the extension points]

- So, instead of creating multiple scheduling processes (pods), just create one scheduling process and attach multiple profiles to it.

```yaml
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler-2
  plugins:
    score:
      disabled:
      - name: TaintToleration
      enabled:
      - name: MyCustomPluginA
      - name: MyCustomPluginB

- schedulerName: my-scheduler-3
  plugins:
    preScore:
      disabled:
      - name: "*"
    score:
      disabled:
      - name: "*"
```

[2 scheduler profiles are there]

- The question arises, if we have multiple pods then we'll have concurrency if the pods chooses different profiles, but in this case there will not be having concurrency.

  ALL THE PROFILES ATTACHED TO A SCHEDULER RUNS IN **MULTIPLE THREADS.**

- The nesting are like the below:

  plugins >> extension point >> disabled/enabled >> ***list of*** **name** (of plugins)

- **Leader Selection**

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
- schedulerName: my-scheduler
  leaderElection:
    leaderElect: true
    resourceNamespace: kube-system
    resourceName: lock-o
```

```
apiVersion: kubescheduler.config.k8s.io/v1
kind: KubeSchedulerConfiguration
profiles:
  - schedulerName: my-scheduler
leaderElection:
  leaderElect: false
```

[have to check if *leaderElection* is scheduler level or profile level]

  - If you enable the leader selection in all schedulers, **only one scheduler** will be active at a time.
  - If you disable the leader selection then **all the schedulers** can be active at same time.
  - Lets say you have 2 schedulers and 4 profiles

    s1: p1 p2

    s2: p3 p4

    and all the profiles are different (i.e. the *schedulerName* are different)

    - **leaderElect: true**

      in this case lets say your pod wants p4 (scheduler profile) and the leader got selected is ***s1***.

      as ***s1*** doesn't contain the required profile i.e. *p4* the pod will remain pending because the selected leader can't schedule the pod.

    - **leaderElect: false**

      in this case, both *s1* and *s2* will be active and in this case s2 (p4) will schedule the pod to specific node.
  - If you keep ***leaderElect: false***, its obvious race condition will occur (if duplicate *schedulerName* are present across the schedulers),

even in that case, **multiple scheduling doesn't happen in kubernetes**. If one scheduler schedule the node, then another scheduler will not be able to do.

How to read from the *docs* ?

- For all the configuration related things like *KubeSchedulerConfiguration*, ..etc etc, you can directly search in the docs.

  kube-scheduler Configuration (v1) - Kubernetes
  Kubernetes › docs › reference › config-api › kube-scheduler-config

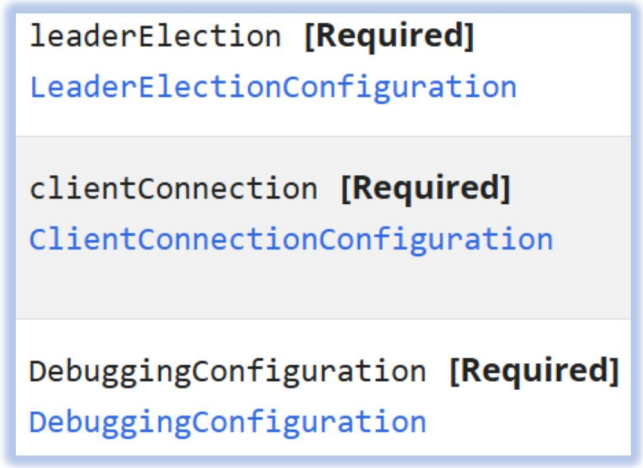
  [you'll get links like this]

- For the *Configuration* (you'll NOT get the docs usnig ***kubectl explain*** command because these are not resources, these are just the configurations), you should go to that ***reference > config-api*** path only (most of the cases).
- Inside that you'll see a lot of things like this.

  [you can see here written: *Appears in: KubeSchedulerConfiguration*]

- Then when you click that link (KubeSchedulerConfiguration)

  you'll see some fields like this.

- It means, the previous field [ClientConnectionConfiguration] values can be written under *clientConnection* field inside *KubeSchedulerConfiguration* .
-

Security Primitives

- RBAC: Roll Based Access Control

  ABAC: Attribute Based Access Control

  Node Authorization

  Webhook Mode

- How to secure the cluster?
  - The core component is API Server. It is the single point of contact for all operations, whether it is via ***kubectl*** or through ***API***
    - Who can access the cluster?
    - Which actions can they perform once access is granted?

Authentication

- In a normal web application backend (what we usually build), the username and password are stored in database and using *JWT* token we verify the user.

  But, Kubernetes doesn't have any database to store the user details. So it depends on a external provider like *static files, certificate authorities, third party identity provider like* ***LDAP*** . ---- for authentication **human users**.

  [Application level user security is managed within the application itself]

- **Static file**
  - You can create a ***csv*** file and write the user details there.

    ```
    password123,user1,u0001,group1
    password123,user2,u0002,group2
    password123,user3,u0003,group2
    ```

    If you want ***tokens***, you can write the same (write token in place of password column)

    ```
    KpjCvBI7rCFAHYPKByTlzRb7gulcUc4B,user10,u0010,group1
    rJjncHmvtXHc6M1WQddhtvNyhgTdxSC,user11,u0011,group1
    mjpOFEiFokLgtoikaRNTt59ePtczZSq,user12,u0012,group2
    ```

    [4th column (group) is completely optional]
  - Then you can mention this file inside the config.

    If *kube-apiserver* is running as a ***systemd*** service (not running inside Pod)

```
ExecStart=/usr/local/bin/kube-apiserver \
  --authorization-mode=Node,RBAC \
  --advertise-address=... \
  --basic-auth-file=user-details.csv
```

if *kube-apiserver* is running inside ***Pod*** then write inside the configuration ***yaml*** file.

```
spec:
  containers:
  - name: kube-apiserver
    image: k8s.gcr.io/kube-apiserver-amd64:v1.11.3
    command:
      - kube-apiserver
      - --authorization-mode=Node,RBAC
      - --advertise-address=172.17.0.107
      - --basic-auth-file=/path/to/user-details.csv
```

- In case of ***username password*** the request will be like the below

  curl -v -k https://master-node-ip:6443/api/v1/pods **-u "user1:password123"**

  in case of ***username token*** the request will be like below

  curl -v -k https://master-node-ip:6443/api/v1/pods **--header "Authorization: Bearer df897ejkhefrt"**

___Generate Certificates for Cluster___

One small Analogy about CA ---------- IMPORTANT --------

- Lets assume CA as Government Authority.
  - It first finds its own identity key [government stamp | *ca.key*]
  - then it creates its own registration form [*ca.csr*]
  - then, it sign that by itself using that stamp [*ca.crt*]
  - [because, the authority has to self sign its own document]
- Now, this government certificate has to be made the source of TRUST so that whatever signed by this will be also a source of TRUST.
- Then, one user came with its details to make it verified.
  - Admin user has its own signature [signature | *admin.key*]
  - it creates a application form like aadhar card details [*admin.csr*]

- Government authority has to approve it with their own stamp

  [*admin.crt*]

  [here, ***ca.key*** was used to sign ***admin.csr*** ; ***ca.crt*** didn't sign but it is the **TRUST** that this certificate belongs to a trusted authority]

- ---

- There are multiple tools that provides certificates like OPENSSL, EASYRSA, CFSSL. We'll use **OpenSSL** here.
- Will start with **CA** (Certificate Authority) certificate [SELF SIGNED]

- All the components that needs Client or Server or Both certificates.
  - Components that acts as *ONLY* Client

    [these all calls the API Server → needs *Client certificate*]
    - kubectl (admin users)
    - kube-scheduler
    - kube-controller-manager
    - kube-proxy
  - Components that acts as *BOTH* Client and Server
    - **kube-apiserver**
      - client cert → talks to etcd, kubelet
      - server cert → serves kubectl / components
    - **kubelet**
      - client cert → talks to API server
      - server cert → API server calls it (exec/logs)
    - **etcd**
      - server cert → serves API server
      - client cert → when it talks peer-to-peer (clustered etcd)
- For the clients to validate certificates sent by server and vice-versa, they all need a copy of ***ca.crt*** (certificate authority).
- Read this Docs to setup SSL for each server and client components.
  https://notes.kodekloud.com/docs/Kubernetes-and-Cloud-Native-Associate-KCNA/Container-Orchestration-Security/TLS-in-Kubernetes-Certificate-Creation/page#generating-server-side-certificates
- Peer-to-Peer ***etcd*** communication: ----------
  - ~~Its not like one etcd server will be there, and another etcd clients.~~

    All are *etcd* ***servers*** only.
  - During communication, one ***etcd server*** will be chosen as the leader, and other will follow them.

    It helps in preventing data inconsistency and duplication.
  - During ***kube-api-server*** configuration, all the ***etcd*** end-points are provided to it [in the ***command*** section if pod, or ***ExecStart*** in case of resource]

- All the **etcd servers** stores the same data [data replicated in all etcd servers]
  if one node dies, then also full data will be available.
- ***etcd servers*** are static (never dynamic), so their end-points are directly provided in *kube-api-server* command.
- 

  [if multiple end-points are there, then *comma separated*]
  ```
  --etcd-servers=https://etcd1:2379,https://etcd2:2379,https://etcd3:2379
  ```
- kube-api-server just connect to one etcd server's end-point,
  etcd cluster handles *leader selection, data consistency, request routing*.
  ~~kube-api-server doesn't find the leader to talk to~~, all that happens internally within ***etcd cluster***.
- F
- F
-

TLS in Kubernetes

- Some high-level concepts:
  - Kubernetes has one ***root authority*** (CA)

    it signs ***client certs, server certs***.
  - Kubernetes uses **mutual TLS** (**mTLS**)

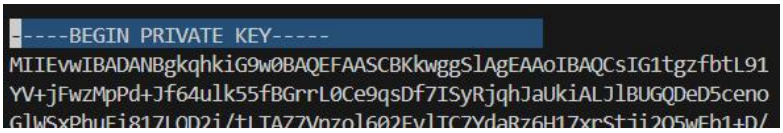
    Client verifies *Server*.

    Server verifies *Client*.

    [that's why both needs certificates]

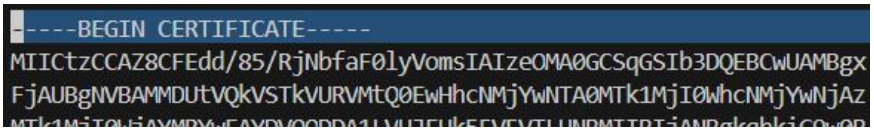
- Generating CA
  -
```
openssl genrsa -out ca.key 2048
openssl req -new -key ca.key -subj "/CN=KUBERNETES-CA" -out ca.csr
openssl x509 -req -in ca.csr -signkey ca.key -out ca.crt
```

    1. create ***ca.key*** [private key (or) *government's stamp*]



    2. create ***ca.csr*** [certificate request (or) *government's self application*]

    3. create ***ca.crt*** (signing with own *key*) [certificate (or) *government's self authorized certificate* (letter of trust)]



    - **CN** : common name

      name of the CA (KUBERNETES-CA in our case)
    - CA self signed its own certificate request because CA is the root. No one is above it.
    - Final output:

      ca.crt, **ca.key**

      [this CA will sign all other certs]

- Creating **Client** Certificates

- Now we'll create certificates for

  admin (kubectl), *scheduler, controller manager, kube-proxy*

- Client Certificate: ***Admin***

```
openssl genrsa -out admin.key 2048
```

```
openssl req -new -key admin.key \
-subj "/CN=kube-admin/O=system:masters" \
-out admin.csr
```

[here, **CN** = username (or) identity (*kube-admin* in here)

O = RBAC group (*system:masters* in here)]

- **system:masters**

  built-in super user group having full access to the cluster, and bypass RBAC restrictions.

  kubectl get clusterrolebindings cluster-admin -o yaml

```
roleRef:
  apiGroup: rbac.authorization.k8s.io
  kind: ClusterRole
  name: cluster-admin
subjects:
- apiGroup: rbac.authorization.k8s.io
  kind: Group
  name: system:masters
```

[**system:masters** group is cluster admin]

- Sign the certificate with CA key and certificate.

```
openssl x509 -req -in admin.csr \
-CA ca.crt -CAkey ca.key \
-out admin.crt
```

- Instead of *username + password*

  certificate + private key [it'll be used]

- Client Certificate: OTHERS
  - Same process as ***admin***, just the **CN** will be changed for others.
  - They'll create *key, certificate request* and

    CA will sign those to generate *certificate*.

- Using Client Certificate
  - Steps:
    - Client sends request
    - Server asks for certificate
    - Client sends admin.crt

- Server:
  - verifies CA
  - extracts CN (common name) and O (organization or group)
- Applies RBAC

- To use the certificates to talk to *kube-apiserver*, you have 2 options.

  kubectl, ***curl*** (with key and certificates)

```
apiVersion: v1
clusters:
- cluster:
    certificate-authority: ca.crt
    server: https://kube-apiserver:6443
  name: kubernetes
kind: Config
users:
- name: kubernetes-admin
  user:
    client-certificate: admin.crt
    client-key: admin.key
```

  [add in the *config* file and use ***kubectl*** to talk to switching to the required ***context***]

```
curl https://kube-apiserver:6443/api/v1/pods \
  --key admin.key --cert admin.crt \
  --cacert ca.crt
```

  [or you can directly pass the *admin key and certificates, CA certificate* with ***curl***]

Creating Server Certificates

Server Certificate: ***etcd***

- ```
  openssl genrsa -out etcd-server.key 2048
  ```

  ```
  openssl req -new -key etcd-server.key \
  -subj "/CN=etcd" \
  -out etcd-server.csr
  ```

  ```
  openssl x509 -req -in etcd-server.csr \
  -CA ca.crt -CAkey ca.key \
  -out etcd-server.crt
  ```

- As ***etcd*** is a distributed database across the cluster, so it needs to talk to the *peer etcd* as well.

  One ***etcdserver.crt***, ***etcdserver.key*** will be there. And remaining will be for *etcdpeer* i.e.
  ```

etcdpeer1.crt, etcdpeer1.key

etcdpeer2.crt, etcdpeer2.key

.... etc

```
- etcd
    - --advertise-client-urls=https://127.0.0.1:2379
    - --key-file=/path-to-certs/etcdserver.key
    - --cert-file=/path-to-certs/etcdserver.crt
    - --client-cert-auth=true
    - --data-dir=/var/lib/etcd
    - --initial-advertise-peer-urls=https://127.0.0.1:2380
    - --initial-cluster=master=https://127.0.0.1:2380
    - --listen-client-urls=https://127.0.0.1:2379
    - --listen-peer-urls=https://127.0.0.1:2380
    - --name=master
    - --peer-cert-file=/path-to-certs/etcdpeer1.crt
    - --peer-client-cert-auth=true
    - --peer-key-file=/etc/kubernetes/pki/etcd/peer.key
    - --peer-trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
    - --snapshot-count=10000
    - --trusted-ca-file=/etc/kubernetes/pki/etcd/ca.crt
```

[it takes the keys and certs like this *(server key & crt, peer key & crt, **etcd CA crt**)* in the *etcd* yaml file]

NOTE: Kubernetes CA is a root, and etcd CA is a root. They don't trust each other.

- Server Certificate: ***kube-apiserver***
  - It's the most important thing because it is the core of Kubernetes cluster that is being used to talk to the whole cluster.
    Either via ***curl*** (direct http REST call) or via ***kubectl***, all the request comes to ***kube-apiserver*** only and it will be delegated from this only.
  - ***kube-apiserver*** is both *server* and *client* component.
    Client: it talks to ***kubelet, etcd***
    Server: serves *kubectl* and other components.
  - It uses 2 different **CA** for its different certificates.
    When it acts as a server (for *kubectl, kubelet, ..etc*), it gets certificate by Kubernetes CA. [*apiserver.crt*]
    When it acts as a client (talking to *etcd*), it gets certificate by ***etcd CA***
    [*apiserver-etcd-client.crt*]

```
--tls-cert-file=/etc/kubernetes/pki/apiserver.crt
--tls-private-key-file=/etc/kubernetes/pki/apiserver.key
--client-ca-file=/etc/kubernetes/pki/ca.crt
```

```
--etcd-cafile=/etc/kubernetes/pki/etcd/ca.crt
--etcd-certfile=/etc/kubernetes/pki/apiserver-etcd-client.crt
--etcd-keyfile=/etc/kubernetes/pki/apiserver-etcd-client.key
```

- Now the question is, how Kubernetes trust 2 different CAs at once?
  - Kubernetes doesn't have a single global trust store.

    Each connection explicitly defines which CA to trust.
  - Trust is not global,

    Trust is **per connection.**
  - ```
    Connection 1 (incoming clients):
        → use Kubernetes CA

    Connection 2 (to etcd):
        → use etcd CA
    ```
- You can see the certificates (client and server) and their respective CAs

```
vagrant@controlplane:/etc/kubernetes/pki$ ls
apiserver-etcd-client.crt  apiserver-kubelet-client.crt  apiserver.crt  ca.crt  etcd                fr
apiserver-etcd-client.key  apiserver-kubelet-client.key  apiserver.key  ca.key  front-proxy-ca.crt  fr
vagrant@controlplane:/etc/kubernetes/pki$ openssl x509 -in apiserver-etcd-client.crt -noout -issuer
issuer=CN = etcd-ca
vagrant@controlplane:/etc/kubernetes/pki$ openssl x509 -in apiserver-kubelet-client.crt -noout -issuer
issuer=CN = kubernetes
vagrant@controlplane:/etc/kubernetes/pki$ openssl x509 -in apiserver.crt -noout -issuer
issuer=CN = kubernetes
```

- **apiserver.crt** --- server crt ----- kubernetes

  apiserver-kubelet-client.crt --- client crt (kubelet) ----- kubernetes

  apiserver-etcd-client.crt --- client crt (etcd) ----- etcd-ca

  [**openssl -in <certificate file> -noout -issuer**] is used to see the CA who signed the certificate.
- API Server is accessed via *kubectl, kubelet, scheduler, controller manager, external users* so it is accessed via

  DNS names, IP addresses, service names

  so the certificate must include all i.e.

  kubernetes, kubernetes.default, kubernetes.default.svc,

  kubernetes.default.svc.cluster.local, cluster IP, node IP
```

- That's why one OpenSSL configuration file has to be created to write the *aliases* (alternate names) and to be used in to create ***apiserver.crt*** certificate.

```
openssl x509 -req -in apiserver.csr \
    -CA ca.crt -CAkey ca.key \
    -out apiserver.crt \
    -extensions v3_req \
    -extfile openssl.cnf
```

```
[req]
req_extensions = v3_req
distinguished_name = req_distinguished_name

[ v3_req ]
basicConstraints = CA:FALSE
keyUsage = nonRepudiation, digitalSignature, keyEncipherment
subjectAltName = @alt_names

[alt_names]
DNS.1 = kubernetes
DNS.2 = kubernetes.default
DNS.3 = kubernetes.default.svc
DNS.4 = kubernetes.default.svc.cluster.local
IP.1 = 10.96.0.1
IP.2 = 172.17.0.87
```

- Server Certificate: ***kubelet***
  - ***kubelet*** will also have 2 kind of certificates, ***client*** and ***server***.
  - **kubelet.crt** and **kubelet.key** are the server certificate and key for kubelet.

    kube-apiserver talks to *kubelet server* to get the details about the pods and send the commands.
  - For all the *kubelet*s as client, they talks to API Server.

    So, for the API Server to know about the *kubelet,* it needs follow a specific naming convention.

    system:node:<nodeName> [this should be the name in the *ssl certificate*]

    for example: *system:node:node01*
  - To verify this, open worker node or controlplane node (kubelet will be there in both if installed by *kubeadm*)

- Execute ***systemctl cat kubelet*** [because *kubelet* doesn't run inside a pod unlike other resources]
  and get the ***config.yaml*** path.

```
Environment="KUBELET_KUBECONFIG_ARGS=--bootstrap-kubeconfig=/etc/kubernetes/b
Environment="KUBELET_CONFIG_ARGS=--config=/var/lib/kubelet/config.yaml"
# This is a file that "kubeadm init" and "kubeadm join" generates at runtime
```

- In that path i.e. */var/lib/kubelet/**pki*** the *certificates and keys* will be present.

```
vagrant@node01:/var/lib/kubelet/pki$ pwd
/var/lib/kubelet/pki
vagrant@node01:/var/lib/kubelet/pki$ ls
kubelet-client-2026-04-30-15-26-19.pem  kubelet-client-current.pem  kubelet.crt  kubelet.key
```

- You can see the *name* and *group* of this *kubelet-client-current.pem* by the command
  openssl x509 -in <file name> ***-noout -text*** [it'll give full details]
  openssl x509 -in <file name> ***-noout -subject*** [it'll give only subjects]

```
vagrant@node01:/var/lib/kubelet/pki$ sudo openssl x509 -in kubelet-client-current.pem -noout -subject
subject=O = system:nodes, CN = system:node:node01
```

  system:node:node01 [node's name is *node01*]
  [due to permission and all, use ***sudo*** with this command]

- F
- F
-

- If you want to see the details of the certificate, then execute this command

  `cat <crt file name> | openssl x509 -text -noout`

```
vagrant@controlplane:/etc/kubernetes$ kubectl config view --raw -o j
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number: 3081854885164004134 (0x2ac4f2b08f80b326)
        Signature Algorithm: sha256WithRSAEncryption
        Issuer: CN = kubernetes
        Validity
            Not Before: Apr 30 15:01:07 2026 GMT
            Not After : Apr 30 15:06:07 2027 GMT
        Subject: O = kubeadm:cluster-admins, CN = kubernetes-admin
        Subject Public Key Info:
            Public Key Algorithm: rsaEncryption
                Public-Key: (2048 bit)
                Modulus:
```

- The below is details about all the servers and clients.

- F

LEGACY NOTES (JUST FOR MORE INFO)

- **Now, we'll generate the client certificates.**
- Admin User (kubectl) [client certificate]
  - Generate admin.key using the below command.

    ```
    openssl genrsa -out admin.key 2048
    ```

  - Generate admin.csr using the below command

    ```
    openssl req -new -key admin.key -subj "/CN=kube-admin" -out admin.csr
    ```

    [the name can be anything, but ***kube-admin*** is the name ***kubectl*** *client* authenticates with]

    if you want to group the keys and certificates (system, server ...etc)

    ```
    openssl req -new -key admin.key -subj "/CN=kube-admin/O=system:masters" -out admin.csr
    ```

    [/O means *Organization* (***group*** in Kubernetes)]

    here the group is **system:masters** , we want a ***admin user***, not a normal user. Keeping the group ***system:masters*** allow the user the admin privileges.

    There is something called **RoleBinding**, in that the *user* is mapped. That user's name must be same as ***/CN*** of the *certificate* otherwise it'll fail.

    Inside the *config file*, the user's name can be different it doesn't create any issue.

    ```
    users:
    - name: system:kube-scheduler
      user:
        client-certificate-data: LS
    ```

    [*/etc/kubernetes/**scheduler.conf***]

    here the user's name can be different, no problem.

    If you see the below, the group name of ***cluster-admins*** is **system:masters** (that's why the group was added to the admin user)

KubeConfig File

- To talk to API Server, you need to attach the *certificates* and all to authenticate yourself.

```
curl https://my-kube-playground:6443/api/v1/pods \
  --key admin.key \
  --cert admin.crt \
  --cacert ca.crt
```

Same for **kubectl** command as well, you have to provide these

```
kubectl get pods \
  --server my-kube-playground:6443 \
  --client-key admin.key \
  --client-certificate admin.crt \
  --certificate-authority ca.crt
```

  - So, rather than writing all these things every time, you can just add the details in ***kubeconfig*** file and just execute the ***kubectl*** command without providing all those keys and certificates.
  - If the *kubeconfig* file is present at ***~/.kube/config*** then ***kubectl*** automatically take that.
    Otherwise, you need to use the flag ***--kubeconfig*** to provide the kubeconfig file's path.

```
kubectl get pods --kubeconfig config
```

- Structure of ***kubeconfig*** file
  - **Clusters**: Represent various Kubernetes clusters (e.g., development, testing, production, or cloud-based clusters).
  - **Users**: Contain credentials and client certificate information for users (e.g., admin, dev, prod).
  - **Contexts**: *Link clusters and users* to define which user accesses which cluster. For example, a context named "admin@production" may indicate that the admin account is used to access the production cluster. It is not necessary that the name will be <user name>@<cluster name>, its just a CONVENTION.

- To change the current context

  kubectl config use-context <context name> [it updates the config file's current-context field]

```yaml
apiVersion: v1
kind: Config
clusters:
- name: my-kube-playground
  cluster:
    certificate-authority: ca.crt
    server: https://my-kube-playground:6443
contexts:
- name: my-kube-admin@my-kube-playground
  context:
    cluster: my-kube-playground
    user: my-kube-admin
users:
- name: my-kube-admin
  user:
    client-certificate: admin.crt
    client-key: admin.key
```

  - Instead of the paths of certificates and key files, you can directly pass the content of those (crt & key) encoded in **base64** format.

    cat ca.crt | base64 [command to encode to base64]

```yaml
apiVersion: v1
clusters:
- cluster:
    certificate-authority-data: LS0tLS1CRUdJT
    server: https://192.168.56.11:6443
  name: kubernetes
contexts:
- context:
    cluster: kubernetes
    user: kubernetes-admin
  name: kubernetes-admin@kubernetes
current-context: kubernetes-admin@kubernetes
kind: Config
users:
- name: kubernetes-admin
  user:
    client-certificate-data: LS0tLS1CRUdJTiBD
    client-key-data: LS0tLS1CRUdJTiBSU0EgUFJ
```

- **To access the API Server using *curl* (direct http request) ---- JUST FOR INFO**
- In my case I couldn't find the ***admin.crt*** and ***admin.key*** files, so I had to convert the *client-certificate-data* and *client-key-data* fields (present inside *~/.kube/config*) to ***base64*** (BEGIN, END) format.

  kubectl config view --raw -o jsonpath='{.users[0].user.client-key-data}' | base64 -d | sudo tee admin.key
  - [**base64 -d** convert the content to *BEGIN, END* type of content]
  - Similarly extract the *client-cert-data* as well.
  - Then just execute the ***curl*** command giving the ***ca certificate***, ***admin certificate***, ***admin key*** and it'll work.

    curl https://192.168.56.11:6443 --cert admin.crt --key admin.key --cacert ca.crt

    [If it doesn't work, then check the permissions of the keys and certificate files]

    [NOTE: It is **https** not *http*]
  - The above base url (https://<IP>:6443) give all the available paths that ***kube-apiserver*** accepts.

  - There are 2 ways you can talk to ***kube-apiserver***, one with ***kubectl*** another is directly ***http request***.
  -

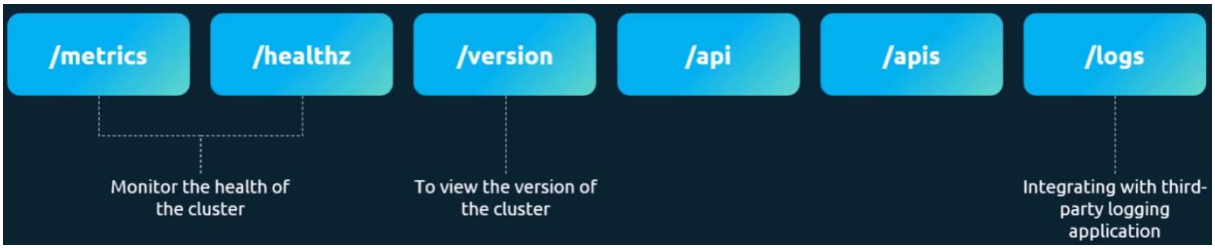
API Server

- When you trigger this http request

  curl https://192.168.56.11:6443 --cert admin.crt --key admin.key --cacert ca.crt

  you'll get the list of *api paths* of the kube apiserver.

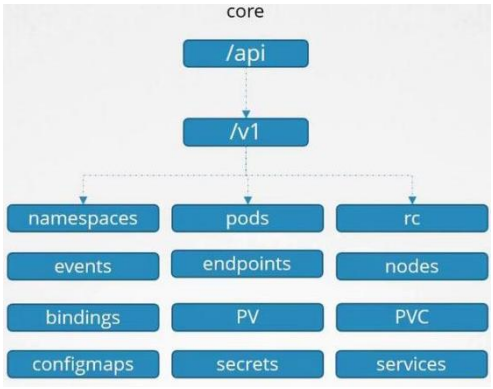
```
vagrant@controlplane:/etc/kubernetes/temp$ sudo curl https://controlplane:6443 --cert admin.crt --key admin.key --cacert ca.crt
{
  "paths": [
    "/.well-known/openid-configuration",
    "/api",
    "/api/v1",
    "/apis",
    "/apis/",
    "/apis/admissionregistration.k8s.io",
    "/apis/admissionregistration.k8s.io/v1",
```

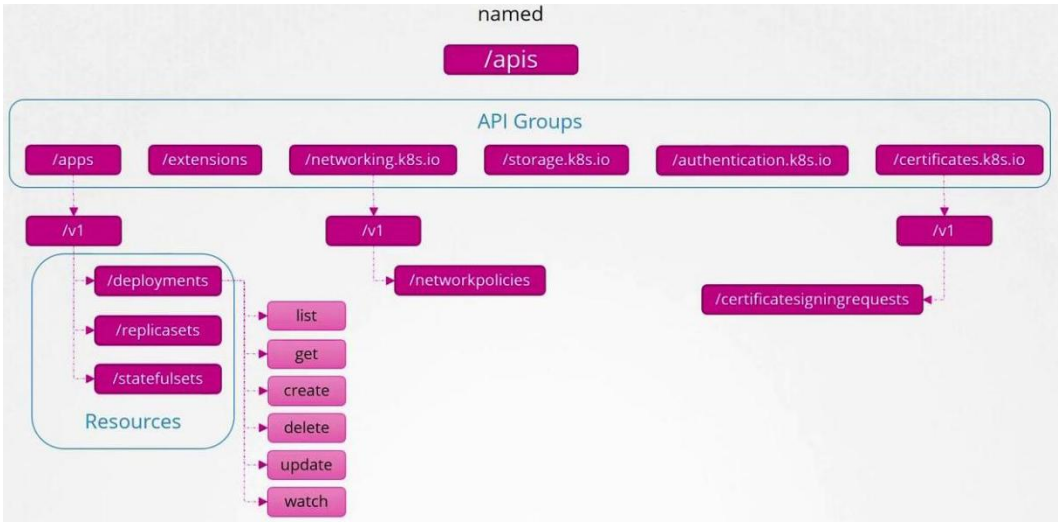


- The cluster functionality is divided in 2 major categories.

  /api : Core Api root Path

  /apis : Named Api root Path





- Whatever things will be added, those will be added in ***named*** API groups because there are properly organized.

  /api is the core API Group.

- **/apis** is named API root path because it contains *API groups* .

  where **/api** doesn't contain any *API group*, rather contains the *version* directly.

- When you write the *yaml* file, you write either "v1" or "<api group name>/<version>" like this.
  - If you write only version i.e. **"v1"** then it'll take the core API root path [**/api**], because it doesn't contain any API group.

    /api/v1/pod, **/api/v1/**services ...etc
  - If you write something like **"apps/v1"** then it'll take the Named API root path [**/apis**] because it contains the groups.

    /apis/apps/v1/replicaset, ..etc
- Names of the components:
  - **/apps**, **/extensions** ...etc present inside ***/apis*** (Named API groups) are called **API Groups**.
  - Inside those **Versions** are there (**v1**).
  - Inside each version, **Resources** are there.

    Ex: ***apps/v1*** contains *deployments, replicasets, statefulsets* ...etc.
  - Each Resources contains **Verbs** like *list, create, delete* ..etc

    ex: *list* the deployments, *create* deployment, *delete* deployment ..etc.

々 To see all the named API groups, execute this command

curl https://controlplane:6443/apis -k --cert admin.crt --key admin.key --cacert ca.crt | grep name [NOTE: *admin.crt, admin.key, ca.crt* should be given]

Authorization

- There are multiple types of Authorization mechanisms
  - Node
  - ABAC [Attribute Based Access Control]
  - RBAC [Role Based Access Control]
  - Webhook
- **Node Authorizer**
  - We saw before, the ***kubelet*** certificate's name was in the form of system:node:<node name>
  - This Node Authorizer is an authorization method of ***kube-apiserver*** that grant permissions to ***kubelets*** (nodes), limit their access to the resources linked with their *respective node only*, prevents nodes from other node's or pod's data.
  - If you see the *subject* of the certificate of *kubelet* in any node, you can see the CN and group

```
vagrant@node01:/var/lib/kubelet/pki$ ls
kubelet-client-2026-04-30-15-26-19.pem  kubelet-client-current.pem  kubelet.crt  kubelet.key
vagrant@node01:/var/lib/kubelet/pki$ sudo openssl x509 -in kubelet-client-current.pem -noout -subject
subject=O = system:nodes, CN = system:node:node01
```

    [CN: system:node:node01, O: system:nodes]

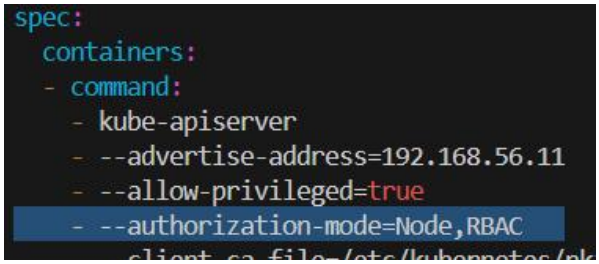
    [node01 is then worker node's name; O means group, CN means common name or name]
  - This certificate is signed by Kubernetes itself.

```
vagrant@node01:/var/lib/kubelet/pki$ sudo openssl x509 -in kubelet-client-current.pem -noout -issuer
issuer=CN = kubernetes
```

  - You **CAN'T CHANGE** this format of system:node:<node name>
    It is HARD-CODED inside Kubernetes.
  - This method is enforced by ***kube-apiserver*** ; it expects the CN and group in that format.
    If the format is not proper, then the request is treated as a *normal user*, NOT a Node.
  - If you see the ***kube-apiserver.yaml*** (Kube Api Server's pod config yaml file)
    [*/etc/kubernetes/manifests/**kube-apiserver.yaml***]
    then you can see the ***--authorization-mode*** flag would have set to **Node**

along with some other authorization mode.

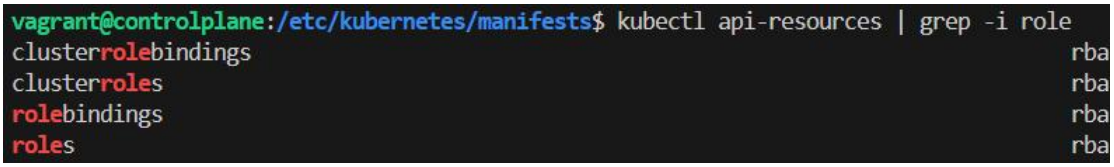
[if kube-apiserver is running as a process, not Pod, then you can see inside ***service*** file using *systemctl cat kube-apiserver*]



- **ABAC** [Attribute Based Access Control]
  - You provide access to a set of users or group a specific set of permissions.
  - In here, you create json file of ***kind: Policy*** and provide that to *kube-apiserver*

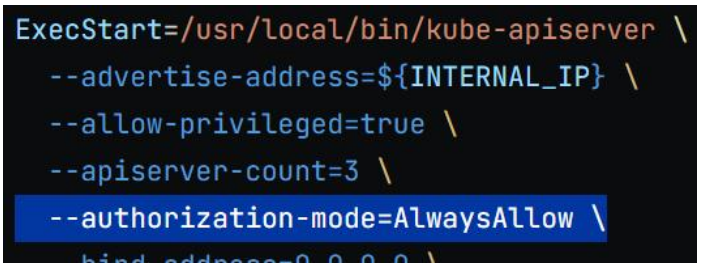
  - When a request hits the kube-apiserver,
    its attributes are extracted:
    user, verb, resources, namespace
    [alok, get, pods, default (For Example)]
    and apiserver checks ABAC Policy file; if match found then *allow* else *reject*.
  - It is ~~not preferable~~ because for each change you need to update the file for each user/group.

- **RBAC** [Role Based Access Control]
  - Instead of writing rules in a file (like ABAC), RBAC uses ***Kubernetes Objects*** [just like pods, replicasets ..etc; you can see in ***kubectl api-resources***],
    you can write *yaml* and create the objects.
  - The object that RBAC provides are
    Role, RoleBinding, ClusterRole, ClusterRoleBinding



- Its kind of, create a role, and associate users/groups to that role.
  If you want to update any permissions, then only need to change in the Role object unlike ABAC where you need to update each Policy.
- Role: permissions within a ***single namespace***.
  ClusterRole: permissions across the whole ***cluster*** or cluster level resources (like nodes)
  - Give access to pod in a namespace: *Role*
    Give access to pod in ALL namespaces: *ClusterRole*
    Give access to nodes: *ClusterRole*
  - **all things that can be done using role can also be done with clusterrole.**
- RoleBinding: grant access within a ***specific namespace***.
  ClusterRoleBinding: grant access across the ***entire cluster***.
  - RoleBinding:
    binds a user/group to ***Role*, or ClusterRole**
    scope: ***one namespace only***
  - ClusterRoleBinding:
    binds a user/group to ***ClusterRole*** *ONLY*
    scope: ***entire cluster***

- **Webhook**
  - It'll make Kubernetes to query an third party REST service to get to know if the user is authorized for that specific request of not.
  - One third party provider is: *Open Policy Client*
- There are 2 more: ***AlwaysAllow***, ***AlwaysDeny***
  - When no authorization mechanism is provided, by default it is *AlwaysAllow*.



- You can give multiple authorization mechanism here (comma separated)

```
ExecStart=/usr/local/bin/kube-apiserver \
  --advertise-address=${INTERNAL_IP} \
  --allow-privileged=true \
  --apiserver-count=3 \
  --authorization-mode=Node,RBAC,Webhook \
```

- When multiple authorization mechanism is given, the execution happens sequentially (from left to right)

  in above case,

  first ***Node*** authorization will happen;

  if it rejects, then RBAC will happen;

  if it rejects, then Webhook will happen.

  - It'll keep trying till the authorization is allowed.

    If it is allowed, then it'll no go to the next authorization mechanism to the right.
  - If all the authorization reject the request, then user will get forbidden.

- To check if you are authorized for doing something, then use **auth can-i** like the below

```
vagrant@controlplane:~$ kubectl auth can-i create deployments
yes
```

- If you want to check the authorization of another user, then use **--as** <user name>

```
vagrant@controlplane:~$ kubectl auth can-i create deployments --as dev-user
no
```

- F

RBAC

- **Role**
  - It is present in "**rbac.authorization.k8s.io/v1**"
  - To create a role, you need to write the rules.
  - Each rule contains *apiGroups, resources, verbs*

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: pod-reader
rules:
- apiGroups:
  - ""
  resources:
  - pods
  verbs:
  - get
  - list
  - watch
```

  - **NOTE**: apiGroups "" doesn't mean ~~all api groups~~, it means Core Api Group (because in core api group is not there)

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: developer
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["list", "get", "create", "update", "delete"]
- apiGroups: [""]
  resources: ["ConfigMap"]
  verbs: ["create"]
```

  - The Role is ***namespace*** *scoped*.

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  namespace: default
  name: pod-reader
rules:
- apiGroups: [""] # "" indicates the core API group
  resources: ["pods"]
  verbs: ["get", "watch", "list"]
```

[**namespace** is provided here]

- Lets say in the *namespace* so many pods are there, but you just want to give access of some specific pods in that role, other pods should not be accssible.

  In that case, you can use ***resourceNames*** field.

```yaml
kind: Role
metadata:
  name: developer
rules:
- apiGroups: [""]
  resources: ["pods"]
  verbs: ["get", "create", "update"]
  resourceNames: ["blue", "orange"]
```

- **RoleBinding**
  - This is to bind a Role to a User/Group.

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: devuser-developer-binding
subjects:
- kind: User
  name: dev-user
  apiGroup: rbac.authorization.k8s.io
roleRef:
  kind: Role
  name: developer
  apiGroup: rbac.authorization.k8s.io
```

  - This is also specific to ***namespace***.
    - NOTE ---------- IMPORTANT ----------

      here you can see **kind: User** [there is nothing called User or Group object (you'll not find this when *kubectl api-resources* is executed.)]

      here, this **subjects[*].kind** is not same as **.kind** (top level); it is just a key to define what kind of subject is that.

      Now, you might be thinking, **apiGroup** is not needed because the *apiGroup* rbac.authorization.k8s.io doesn't contain anything called

User or *Group* because if it contains, then it would be a Kubernetes Object (like pod, replicaset, role, clusterrole, ...etc)

kind can be *User, Group* or *ServiceAccount* ; and *ServiceAccount* is a real object (present in Core API Group) but *User* and *Group* are not object.

The RBAC matching logic is written using the combination of **kind** and **apiGroup**, so if you don't give *apiGroup* it'll fail.

`(kind + apiGroup)`

it'll be always

`apiGroup: rbac.authorization.k8s.io` (for User, Group) or

`apiGroup: ""` (for ServiceAccount)

```yaml
apiVersion: rbac.authorization.k8s.io/v1
# This role binding allows "jane" to read pods in the "default" namespace.
# You need to already have a Role named "pod-reader" in that namespace.
kind: RoleBinding
metadata:
  name: read-pods
  namespace: default
subjects:
# You can specify more than one "subject"
- kind: User
  name: jane # "name" is case sensitive
  apiGroup: rbac.authorization.k8s.io
roleRef:
  # "roleRef" specifies the binding to a Role / ClusterRole
  kind: Role #this must be Role or ClusterRole
  name: pod-reader # this must match the name of the Role or ClusterRole you wish to bind to
  apiGroup: rbac.authorization.k8s.io
```

One confusion is **namespace**, if we have given *namespace* field inside ***Role***, then why do we need to give it again in ***RoleBinding***?

namespace in ***Role*** defines "In which namespace the Role should be created"

namespace in ***RoleBinding*** defines "I'll bind to the role present in that *namespace*"

so basically, if you have given ***namespace: ns1*** in the *RoleBinding*, then

the specific role must be present in the ***namespace ns1***. if the role is present in some other namespace then it'll not work.

- [its only for **Role**. Not for **ClusterRole**]

- **Role** and **RoleBinding** are *namespace scoped*.

  If ***namespace*** is not provided, then the objects (*Role, RoleBinding*) will be created in the same *namespace* where they are being called.

  Not the ~~*default* namespace~~.

- **ClusterRole**
  - It is same as ***Role***, but it is not limited to any *namespace*.

    Its ***cluster scoped***.

```yaml
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  # "namespace" omitted since ClusterRoles are not namespaced
  name: secret-reader
rules:
- apiGroups: [""]
  #
  # at the HTTP level, the name of the resource for accessing Secret
  # objects is "secrets"
  resources: ["secrets"]
  verbs: ["get", "watch", "list"]
```

  - unlike *Role*, ***namespace*** field will not be here, because it'll be available ***cluster wide***.
  - One more difference between Role and ClusterRole is:

    Role can have ***namespace*** resources.

    ClusterRole can have both ***namespace*** and ***cluster-scoped*** resources.
  - To check namespace and non-namespaced resources, execute the command

    kubectl api-resources --namespaced=true [namespaced resources]

    kubectl api-resources --namespaced=false [non-namespaced

resources]

```
vagrant@controlplane:~$ kubectl api-resources --namespaced=false
NAME                              SHORTNAMES                                  APIV
componentstatuses                 cs                                          v1
namespaces                        ns                                          v1
nodes                             no                                          v1
persistentvolumes                 pv                                          v1
mutatingadmissionpolicies                                                     admi
mutatingadmissionpolicybindings                                               admi
mutatingwebhookconfigurations                                                 admi
validatingadmissionpolicies                                                   admi
```

- **ClusterRoleBinding**
  - It is used to bind the ***ClusterRole*** to any User or Group or ServiceAccount.
  - If you bind a *ClusterRole* using ***ClusterRoleBinding*** then the authorized resources (mentioned in *ClusterRole*) will be authorized to the User or Group or ServiceAccount with which the *ClusterRole* is bound to.
  - ***ClusterRole*** can be bound with ***RoleBinding*** as well, in this case the *authorization* will be applied to a specific namespace where ***RoleBinding*** will be created.
- After creating a binding, if you want to change the ***roleRef*** you'll get a validation error.

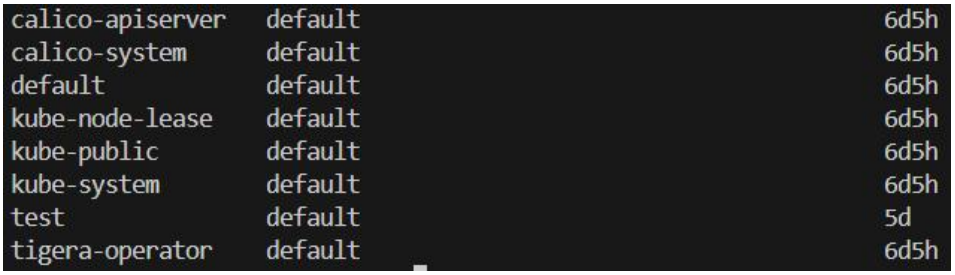
  If you really want to do it, then you need to delete the *binding* object and create a new one with the new *roleRef*.
-

ServiceAccount

- There are 2 types of accounts: **User, Service**.
  - User account is used by the users like admin, developer.
  - Service account is used by applications to interact with kubernetes cluster like *Jenkins, Prometheus* ..etc.

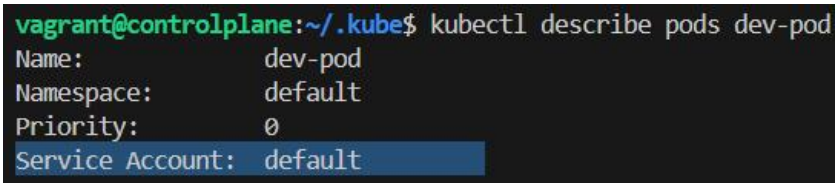
    [Prometheus pulls kubernetes api for the performance matrix]

    [Jenkins used Service Account to deploy the application in Kubernetes cluster]

- In all namespace, there will be a default *ServiceAccount* whose name is **"default"**.
  - 

    ```
    calico-apiserver   default   6d5h
    calico-system      default   6d5h
    default            default   6d5h
    kube-node-lease    default   6d5h
    kube-public        default   6d5h
    kube-system        default   6d5h
    test               default   5d
    tigera-operator    default   6d5h
    ```

    [namespace ServiceAccount AGE] (columns)
  - When a pod is created, by-default the *ServiceAccount* present in that namespace (where pod is created) will be attached to the pod *IF NOT EXPLICITLY MENTIONED*.

    ```
    vagrant@controlplane:~/.kube$ kubectl describe pods dev-pod
    Name:             dev-pod
    Namespace:        default
    Priority:         0
    Service Account:  default
    ```
  - **Pod needs the ServiceAccount to talk to API Server.**

    [there are multiple reasons for which Pod talks to API Server.

    But, ~~pod-to-pod communication doesn't happen through API Server~~]

- Both ***internal components*** (like pods, ..etc) and *external services* (like Jenkins ..etc) uses **ServiceAccount** to communicate with API Server.
  - ServiceAccount is required to give access to the resources via ***Role*** or ***ClusterRole***.
  - For external services, you need to create a token linked to the ServiceAccount.

    With that token (Authorization Bearer) the external service can

communicate to API server.

kubectl create token <service account name>

```
kubectl create token external-sa -n dev
```

NOTE: Token **doesn't have name**, and it is **not stored anywhere**.

When you execute this command, a token will be generated and displayed in the screen using which the Service can trigger REST request using

Bearer <generated token>.

- For internal components (like Pod), the *Service Account* is directly attached to it. But that doesn't mean it doesn't require token.

  For internal components, a token is generated automatically using which they can talk to API Server.

- When a *Service Account* is attached to a Pod, kubernetes creates a virtual volume and mounts that to the Pod. [if

  spec.automountServiceAccountToken: true inside Pod

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-kubernetes-dashboard
spec:
  containers:
    - name: my-kubernetes-dashboard
      image: my-kubernetes-dashboard
  automountServiceAccountToken: false
```

  (or)

  automountServiceAccountToken: true inside ServiceAccount]

  This mount path is:

  /var/run/secrets/kubernetes.io/serviceaccount/token [inside Pod]

```
Mounts:
  /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-rdvhr (ro)
```

  this contains *token, ca.crt, namespace* details.

  [NOTE: volume's lifecycle = pod's life cycle; if pod is gone, volume is also destroyed]

- In short,

  ServiceAccount is required to provide specific authorization in

Role or ClusterRole.

After that, a token will be generated linked to that Service Account (automatic for internal components, manually for external services) and that token is not stored anywhere.

For Pods, the token is generated and stored inside that mounted path and it gets refreshed when it is expired.

Virtual volume’s is destroyed once pod is gone.

Image Security

- All the official images like nginx/nginx, redis/redis ..etc will be having a verified badge “Docker Official Image”.

  these all are official images maintained by Docker.

  When you execute **docker pull nginx**,

  it pulls from **library/nginx** (library stores all the official images)

- When you provide the image name, it actually translates that to **library/**<image name>

  because it is where the official images are present in docker hub.

  ```
  image: library/nginx
  ```

- It pulls from Docker Registry (Docker Hub) by default,

  ```
  image: docker.io/library/nginx
  ```

- But, if you are having a private container registry and want to pull the image from there, then you need to give full path against the *image* key.

  ```
  image: private-registry.io/apps/internal-app
  ```

- Now, as this is private registry and it needs to be authenticated to access that, you need to create a resource called **secret** .

  ```
  kubectl create secret docker-registry regcred \
    --docker-server=private-registry.io \
    --docker-username=registry-user \
    --docker-password=registry-password \
    --docker-email=registry-user@org.com
  ```

  docker-registry: type of secret

  regcred: name of secret (just an identifier)

- And mention that *secret* in the Pod's yaml file inside the field ***spec.imagePullSecrets***

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: nginx-pod
spec:
  containers:
  - name: nginx
    image: private-registry.io/apps/internal-app
  imagePullSecrets:
  - name: regcred
```

```yaml
spec:
  imagePullSecrets:
  - name: my-registry-secret-1
  - name: my-registry-secret-2
```

- *pod.spec.**imagePullSecrets*** is a list; it'll try one by one until it finds the required image.

  [containers and imagePullSecrets: both are siblings]

- Create **secret** from yaml

```yaml
apiVersion: v1
data:
  .dockerconfigjson: eyJhdXRocyI6eyJ
kind: Secret
metadata:
  name: my-secret
type: kubernetes.io/dockerconfigjson
```

[yaml file looks something like this]

  - *secret.data.\.dockerconfigjson* (\. means to keep . as a string)
  - That is **base64** encoded.

    On decoding that:

```yaml
{
  "auths": {
    "DOCKER_REGISTRY_SERVER": {
      "username": "DOCKER_USER",
      "password": "DOCKER_PASSWORD",
      "email": "DOCKER_EMAIL",
      "auth": "RE9DS0VSX1VTRVI6RE9DS0VSX1BBU1NXT1JE"
    }
  }
}
```

[something like this will come]

-

Security Context

- In docker, while creating a container, you can specify if you don't want the ***root*** user as default.

  [root is risky, if Pod is compromised, then it'll have full access because of root user]

  you can specify ***--user=***<user id> while running Docker container.

  docker run --user=1001 ubuntu

- When a container runs, it doesn't get full root privilages *even if it is root user*.

  Docker uses a Linux feature called ***capabilities*** to split root power into smaller pieces.

  Using ***--cap-add*** <capability name> you can provide the specific capability to the user.

  docker run --cap-add MAC_ADMIN ubuntu

  `NET_ADMIN` → control networking
  `SYS_ADMIN` → system-level control
  `CHOWN` → change file ownership
  `MAC_ADMIN` → control security policies

- In Kubernetes also you can configure all of these.
  - As, Containers run inside a Pod in Kubernetes.

    So, if you configure the security settings at Pod level, it'll be **applied to all the containers** running inside it.

    If you configure the security settings at both Pod and Container level, Pod's security settings will be **overridden by Container's security settings**.

- You can add a field ***spec.containers[*].securityContext*** to specify the security configurations *at container level*.

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: web-pod
spec:
  containers:
    - name: ubuntu
      image: ubuntu
      command: ["sleep", "3600"]
      securityContext:
        runAsUser: 1000
        capabilities:
          add: ["MAC_ADMIN"]
```

[Container Level]

- If you want to configure it in ***pod level*** then, add the field ***spec.securityContext*** .

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: web-pod
spec:
  securityContext:
    runAsUser: 1000
    capabilities:
      add: ["MAC_ADMIN"]
  containers:
```

々 F

Network Policies

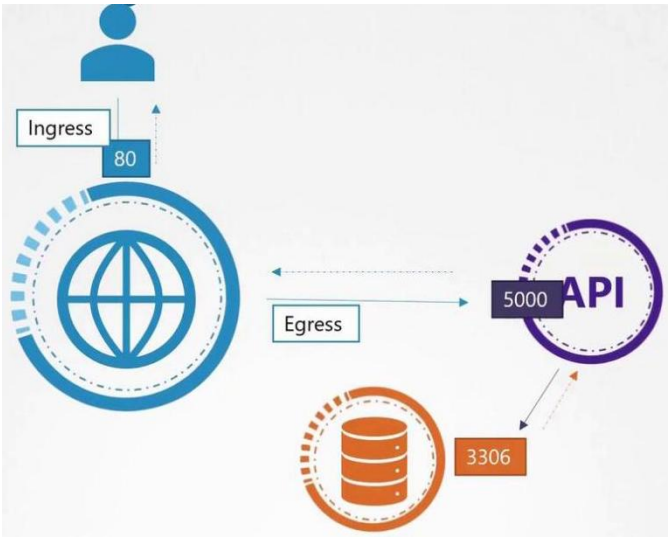
- In Kubernetes, there is no restriction for inter Pod communication by default.

  Any pod can talk to any other pod using the target Pod's service (ClusterIP).

  - To restrict this *Network Policies* are used.
  - Its kind of ABAC (Attribute Based Access Control), because here also Policy was there which had to be written in a JSON format.

    [Not same as ABAC, because ABAC is of course used for Authorization; and Network Policy is used for restrict the access of Pods]
  - For example, there are 3 types of pods running.
    - ***Frontend, Backend, DB***
    - By default, Frontend can access the DB pods which should not happen.

- Consider the below image

  - Web Server:

    Ingress: accept HTTP traffic on Port 80

    Egress: sends traffic on port 5000 to Backend API server.
  - Backend API Server

    Ingress: Port 5000

    Egress: Port 3306 to DB
  - DB

    Ingress: Port 3306

- **NetworkPolicy**
  - It is **namespace scoped**.
  - controls traffic to and from pods within that namespace, optionally including traffic from other namespaces via selectors

```yaml
apiVersion: networking.k8s.io/
kind: NetworkPolicy
metadata:
  name: test-network-policy
  namespace: default
spec:
  podSelector:
    matchLabels:
      role: db
  policyTypes:
  - Ingress
  - Egress
  ingress:
  - from:
    - ipBlock:
        cidr: 172.17.0.0/16
        except:
        - 172.17.1.0/24
    - namespaceSelector:
        matchLabels:
          project: myproject
    - podSelector:
        matchLabels:
          role: frontend
    ports:
    - protocol: TCP
      port: 6379
  egress:
  - to:
    - ipBlock:
        cidr: 10.0.0.0/24
    ports:
    - protocol: TCP
      port: 5978
```

  - **spec.podSelector** selects the pod in which NetworkPolicy will be applied. (here **db**)
  - **spec.ingress** and **spec.egress** contains all the configs (which pod's traffic at which port will be allowed)
  - What is the use of **spec.policyTypes** then?

    It enforces the ingress and egress rules.
    - If *policyTypes* field is not there, Kubernetes will add the values (Ingress, Egress) based on presence of *networkpolicy.spec.ingress* and *networkpolicy.spec.egress* fields.

      If only ingress is present, it becomes Ingress; if only egress is present, it becomes Egress; if both are present, it becomes both

Ingress and Egress.

In the live configuration (**kubectl get networkpolicy** <name> **-o yaml**), the *policyTypes* field will always be present, either defined by the user or added by Kubernetes.

- In the live config, whatever values are there [Ingress or Egress or both] inside *networkpolicy.spec.policyTypes*, only those will be enforced.

  If a direction [Ingress or Egress] is listed in policyTypes but no corresponding rules are defined, then that direction becomes completely blocked (**default deny**).

  For example, if policyTypes contains Ingress but no ingress rules are defined, then all incoming traffic is blocked.

- **namespaceSelector**
  - if provided, namespaces with those labels will be selected.
  - If not provided, pods within the same namespace (not all namespaces) will be allowed.

- F

Cluster Networking

- In a Kubernetes cluster, all the nodes must be having at least one network interface configured with
  - Unique IP address
  - Unique hostname
  - Unique MAC address.
- **hostname** is system level, not specific to a network interace.
  - But, in ***/etc/hosts*** file, the IP and hostnames are written.

    ```
    192.168.56.11 controlplane
    192.168.56.21 node01
    192.168.56.22 node02
    ```

    [in *controlplane* node]
  - In that controlplane node (where this /etc/hosts file is present), if I specify **node02** then it'll take the linked IP only which is *192.168.56.22*
  - In pure networking, if the node having IP 192.168.56.22 (which name is written as *node02* in /etc/hosts of controlplane) is having some different hostname (let say ***n2***) then also it should work.
    Because anyhow the IP is being extracted from the controlplane node only while writing *node02*.
  - But, In Kubernetes it'll break the things.
- Some ports needs to be accessible on the node.
  [any port can be called from a port. But the ***firewall*** must let the incoming traffic in]
  - **Control plane nodes**

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	6443	Kubernetes API server	All
TCP	Inbound	2379-2380	etcd server client API	kube-apiserver, etcd
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10259	kube-scheduler	Self
TCP	Inbound	10257	kube-controller-manager	Self

- **Worker nodes**

Protocol	Direction	Port Range	Purpose	Used By
TCP	Inbound	10250	Kubelet API	Self, Control plane
TCP	Inbound	10256	kube-proxy	Self, Load balancers
TCP	Inbound	30000-32767	NodePort Services†	All
UDP	Inbound	30000-32767	NodePort Services†	All

  - Why do the ports of *controller manager, scheduler* are opened because they are never called by others, instead they call api servers?

    They exposes ports for health checks, monitoring, logging ..etc.
  - For etcd, 2 ports are mentioned: *2379* & *2380*

    2379: client traffic (kube api server, etcdctl ..etc)

    2380: peer communication (used by other etcd nodes)

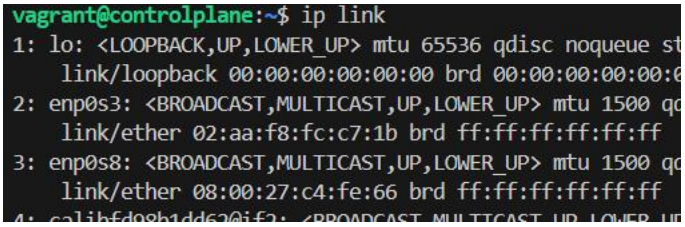
- Networking internals:
  - Cluster requires to allocate non-overlapping IPs to nodes, services and pods.

    Network plugin (CNI) allocate IP to Pods.

    API Server allocate IP to Services.

    Kubelet or cloud-controller-manager (CCM) allocate IP to Nodes.
- Some important commands
  - **ip link**

    displays network interfaces (eth0, eth1 ..), MAC addresses, interface state (UP/DOWN)



- **ip addr** [legacy one is: *ifconfig*]

  displays IP addresses assigned to interfaces (etho 192.168.1.0)

```
vagrant@controlplane:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue sta
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdi
    link/ether 02:aa:f8:fc:c7:1b brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope glo
       valid_lft 80720sec preferred_lft 80720sec
    inet6 fd17:625c:f037:2:aa:f8ff:fefc:c71b/64 scope glo
       valid_lft 85921sec preferred_lft 13921sec
```

- **ip addr add** 192.168.1.10/24 **dev** etho

  assigns IP address to an interface. [here etho --> 192.168.1.10]

- **ip route** [legacy one is: *route -n*]

  shows routing table

  ex: *default via 192.168.1.1*

```
vagrant@controlplane:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp s
10.0.2.0/24 dev enp0s3 proto kernel scope li
10.0.2.2 dev enp0s3 proto dhcp scope link sr
blackhole 10.244.49.64/26 proto 80
10.244.49.65 dev calibfd98b1dd62 scope link
10.244.49.66 dev cali93d2b2b0737 scope link
10.244.49.67 dev cali0240531d65d scope link
10.244.49.68 dev calia26e734911e scope link
10.244.49.69 dev cali0970f975c8a scope link
10.244.49.70 dev cali5fdb66d5490 scope link
10.244.49.71 dev calife997dd1838 scope link
10.244.49.72 dev calif1ce4f5f5f2 scope link
10.244.140.64 via 192.168.56.22 dev enp0s8 p
10.244.140.64/26 via 192.168.56.22 dev enp0s
```

- **ip route add** 192.168.1.0/24 **via** 192.168.2.1

  adds a route [here, to reach 192.168.1.x, go via ***gateway*** 192.168.2.1]

- **arp**

  displays ARP table (IP -> MAC mapping)

- **route**

  older version of *ip route* command.

- **netstat -plnt**

  shows listening ports and processes.

  [ex: which service is running at port 6443]

- cat /proc/sys/net/ipv4/ip_forward

  checks if IP forwarding is enable.

  0: disable; 1: enable

々 F

Pod Networking

- [pre-requisite; Linux Namespace & bridge networking]
- Consider the below scenario:

Node	Host IP	Pod CIDR	Bridge IP	Pods
Node-1	192.168.1.11	10.244.1.0/24	10.244.1.1	10.244.1.2, 10.244.1.3
Node-2	192.168.1.12	10.244.2.0/24	10.244.2.1	10.244.2.2
Node-3	192.168.1.13	10.244.3.0/24	10.244.3.1	10.244.3.2

- To create these
  - create a **bridge** on every node and assign it a IP (here 10.244.1.1, 10.244.2.1 ..etc) [**Bridge IP**]
  - Create veth-pair (one per pod) and connect pod and bridge in the respective nodes.
  - Assign IP to each pods in 10.244.1.x, 10.244.2.x ..etc. [**Pod IPs**]
    So, Pod CIDR (Bridge's network) will be 10.244.1.0/24, 10.244.2.0/24, ..etc.
- To make all pods communicate with each other across nodes.
  - Configure default gateway inside each pods (Pod namespace)
    10.244.x.1 [x: 1, 2, 3 .. etc for respective nodes]
    [bridge's IP]
    - Means, if IP address is out of its subnet, then it'll go via *bridge*.
  - Update **gateway** in all the **nodes** (not pods).
    i.e.
    network: 10.244.1.0/24, gateway: 192.168.1.11
    network: 10.244.2.0/24, gateway: 192.168.1.12
    network: 10.244.3.0/24, gateway: 192.168.1.13
    - Now, each node knows where to send the packet to reach the respective pods.
  - Enable IP forwarding on each Node. Because for going out of the node,
    --- just example ---
    src IP: 10.244.1.2 (node-1 pod)
    dest IP: 10.244.2.2 (node-2 pod)

- For this, it should go from 10.244.1.**1** (node-1 bridge) to 192.168.1.11 (node-1's LAN ip)
- To forward packet from one network (10.244 ...) to another (192.168. ..) or *vice-versa*, **node-1 OS** acts as a router here.
  So, IP forwarding needs to be enabled.

- NOW, PACKET FORWARD CAN HAPPEN SMOOTHLY.
  Pod can reach its host's LAN port (default gateway is bridge, and IP forwarding inside host)
  from host to another node's pod (pod's gateways are their respective nodes)
  another node can send packet to its pod (private network inside it only and IP of bridge is configured)
  that pod can now see the src address (host's pod) and send response back via its respective bridge -> node -> host -> bridge -> host's pod.

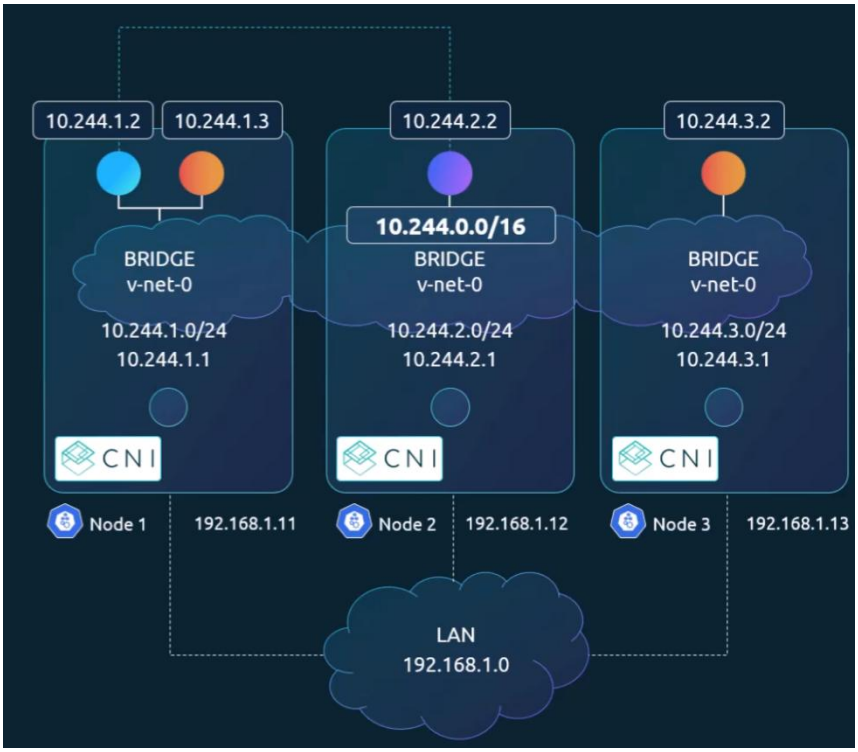
- In the previous approach, we were writing the *gateways* in all the nodes which is not maintainable in large scale setup.
  - Connect all the nodes to a router [IP: **192.168.1.1**]
  - Add router's IP as the default gateway in all the nodes.
    [so for every pod's IP, node will not able to find the gateway, so they'll forward packet to default gateway which is their router]
  - Router will have all the route gateways (which we were writing in all the nodes)

Network	Gateway
10.244.1.0/24	192.168.1.11
10.244.2.0/24	192.168.1.12
10.244.3.0/24	192.168.1.13

    all at one place.
  - Now, router will forward the packet to respective node;
    and that node to its respective pod.
  - now, the individual pod namespaces will become a single entity throughout the cluster having network ip
    10.244.0.0./16

[*10.244.1.0/24 -- node-1's pod namespaces*;

10.244.2.0/24 -- node-2's pod namespaces] ... etc



[individual namespace became cluster wide namespace now having subnet **10.244.0.0/16**]

ConfigMap

- Kubernetes object [apiVersion: **"v1"**] to keep env variables which can be used
  - *Inside a container's command (cmd) and args*.
  - *Environment variables for a container*.
- Inside a container's command (cmd) and args
  -
```
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config

data:
  app_mod: production
```
[ConfigMap]

    here the variables are written under *ConfigMap.**data*** field (not inside *spec* unlike others)
    - app_mod: production (this is the only key value pair written here which can be used)
  -
```
apiVersion: v1
kind: Pod
metadata:
  name: demo-pod
spec:
  containers:
    - name: app
      image: busybox
      command: ["/bin/sh"]
      args: ["-c", "echo Running in $APP_MOD; sleep 3600"]
      env:
        - name: APP_MOD
          valueFrom:
            configMapKeyRef:
              name: app-config
              key: app_mod
```
[Pod]

    - Here pod.spec.containers[*].*args* is using ***APP_MOD*** (Upper case) But, in ConfigMap it was ***app_mod*** (Lower case)

- In pod.spec.***env***

  name: **APP_MOD**

  and this is derived from *ConfigMap.**app_mod***

  - Inside Pod, **APP_MOD** (upper case) has value of **pod_map** (of ConfigMap)

- So, inside pod.spec.containers[*].**args** is using **APP_MOD** instead of ~~**app_mod**~~ .
- Commands and args are anyways combined:

  ```
  command: ["/bin/sh", "-c", "echo Running in $(APP_MOD); sleep 3600"]
  ```

  this is same as the previous one

- Environment variables for a container
  - ```
    apiVersion: v1
    kind: ConfigMap
    metadata:
      name: db-config
    data:
      DB_HOST: mysql-service
      DB_PORT: "3306"
    ```

    [ConfigMap]
  - ```
    apiVersion: v1
    kind: Pod
    metadata:
      name: env-pod
    spec:
      containers:
        - name: app
          image: nginx
          env:
            - name: DATABASE_HOST
              valueFrom:
                configMapKeyRef:
                  name: db-config
                  key: DB_HOST

            - name: DATABASE_PORT
              valueFrom:
                configMapKeyRef:
                  name: db-config
                  key: DB_PORT
    ```

    [Pod]

- If you execute the below inside the container

```
echo $DATABASE_HOST
echo $DATABASE_PORT
```

  then you'll get the values set inside ConfigMap.
- NOTE: the ConfigMap's keys are DB_HOST, DB_PORT but in the containers config (in pod yaml config) the names are DATABASE_HOST, DATABASE_PORT
  so this should be used.
-

々 F

々 F

々 F

々

々 F

々 F

々

CNI in Kubernetes

- Container runtime (containerd, cri-o) calls the cli plugins to assign ips and all to the pods.
  - There are 2 main directories

    /etc/cni/net.d/

    /opt/cni/bin/
  - **/etc/cni/net.d/**
    - This contains CNI Network Configuration files.
    - It tells kubelet/container runtime about

      which CNI plugin to use.

      And how to configure pod networking.
    - Files can be of extension **.conf, .json, .conflist**
    - Ex: 10-calico.conflist, 10-flannel.conflist, 10-weave.conflist

      ```
      root@controlplane:~# ls /etc/cni/net.d/
      10-calico.conflist  calico-kubeconfig
      ```
    - If multiple config files are there, it choose in alphabetical order.
    - It just contains configurations about which plugin to use and with what configurations.

      It doesn't contains any executables.
    - *type* : which CNI plugins to execute (calico, flannel, ..etc)

      ipam (ip add management) : how pod IP addresses will be allocated?

      name : logical name of CNI network (just an identifier)

      k8s api details : how CNI plugin communicate with k8s api server

      plugins : (list) run multiple plugins one after another

      mtu (maximum transmission unit) : maximum packet size allowed

      port mapping settings : allow container ports to be exposed on host.
  - **/opt/cni/bin/**
    - This directly contains actual CNI plugin executables.

      ```
      -rwxr-xr-x 1 root root  5698763 Apr 24 10:38 bridge
      -rwxr-xr-x 1 root root 79605982 Apr 30 15:14 calico
      -rwxr-xr-x 1 root root 79605982 Apr 30 15:14 calico-ipam
      -rwxr-xr-x 1 root root 13725422 Apr 24 10:38 dhcp
      -rwxr-xr-x 1 root root  5251069 Apr 24 10:38 dummy
      -rwxr-xr-x 1 root root  5702145 Apr 24 10:38 firewall
      -rwxr-xr-x 1 root root  2862657 Apr 30 15:14 flannel
      ```

- These binaries can *create veth pairs, create bridges, assign pod IPs, configure routes, configure iptables, configure NAT, connect pod namespace, setup overlays, enforce policies*.

- So basically

  /etc/cni/net.d : instructions

  /opt/cni/bin : actual workers

Ingress

- Consider the below scenario:
  - You have a deployment having multiple replicas.
    To access them you created a NodePort service which is exposed at port *30080*.
    let the service name is: *wear-service*
  - Now, you want the users to access it via a domain, but the port here is 30080 (NodePort's port must be >=30000).
    so, users can access it via
    http://my-online-store.com:30080
  - But, this is not proper. User should not enter the port directly.
    But to avoid port, it needs to be a valid HTTP (80) or HTTPS(443) port.
    [http://.... will use *80* port, https://.... will use *443* port]
  - So, now you need to add a reverse proxy (load balancer) outside of the cluster whose IP will be added in the domain and the traffic can be routed properly with the url
    http://my-online-store.com
  - If you are deploying in a public cloud environment like AWS, Azure, GCP, then instead of creating a service of type *NodePort*, you can create a service of type *LoadBalancer*.
    It'll create a NodePort and also a Load balancer in the cloud by itself.
  - Now, you are deploying one more types of pods, and a service for those.
    Let say the service name s: *watch-service*
    and you want it to be accessed at
    *http://my-online-store.com/**watch*** or *http://**watch.**my-online-store.com*
  - Now this service will also be of type *LoadBalancer*, so the cloud platform will create another load balancer and NodePort service for this new deployment.
  - So for each new deployment, one more load balancer will be created.
  - Now all the load balancers will have separate IPs, and to direct the request to the respective load balancers, you again need to create a Load Balancer.

/wear : load-balancer-1

/watch : load-balancer-2

- Each time a new deployment comes, one new load balancer has to be created.

  Then the main load balancer will have to be configured again.

  And multiple load balancer will cost more.

- How to fix this issue?
  - You can ask the developers to handle these at application level. (/wear, /watch routes)

    but this is not the right way. Because, it should not be done at application level as the features might be completely different.
  - In this case you can use Ingress,
- Ingress
  - It's a *layer 7 load balancing solution* that provides a single end-point that routes the traffic to specific services depending upon the path or url user has provided.
  - It can be configured like

    my-online-store.com/wear, my-online-store.com/watch : it'll go to respective wear or watch services [path based]

    wear.my-online-store.com, watch.my-online-store.com [host based]
  - It also provides SSL security, load balancing along with HTTP routing.
  - But still you need to expose the Ingress port with a NodePort or LoadBalancer service.

    But this is just a one-time effort. Even if a new deployment comes, that will be handled at Ingress only without any need of changing the load balancer or node port service.
- Service : forwards request to a Pod

  Ingress : forwards request to a Service
- Ingress in Kubernetes
  - Kubernetes doesn't provide any Ingress controller by default.

    You need to deploy by your own which will route the traffic to a specific service.
  - Multiple options are there like *nginx-ingress, HAPROXY, traefik* ..etc.

- Then you need to maintain a config file which contains set of rules about the traffic routing.

  Its called Ingress Resources.
- ***Ingress Resources***: it defines the rule about what to do and how.

  Ingress Controllers : it performs the actions according to those rules.
- nginx provide load balancing.

  But in **nginx-ingress** (Ingress Controller): load balancing is just a part of it. Along with it, it has built-in additional intelligence to monitor kubernetes cluster for new definitions or ingress resources and configure the nginx server accordingly.
- *nginx-ingress* is deployed in a Deployment just like others in a Pod.

Configuring **Ingress Controller**

- First create a deployment of *nginx-ingress* , keeping replicas=1 only.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-ingress-controller
spec:
  replicas: 1
  selector:
    matchLabels:
      name: nginx-ingress
  template:
    metadata:
      labels:
        name: nginx-ingress
    spec:
      containers:
        - name: nginx-ingress-controller
          image: quay.io/kubernetes-ingress-controller/nginx-ingress-
                 controller:0.21.0
          args:
            - /nginx-ingress-controller
            - --configmap=$(POD_NAMESPACE)/nginx-configuration
          env:
            - name: POD_NAME
              valueFrom:
                fieldRef:
                  fieldPath: metadata.name
            - name: POD_NAMESPACE
              valueFrom:
                fieldRef:
                  fieldPath: metadata.namespace
          ports:
            - name: http
              containerPort: 80
            - name: https
              containerPort: 443
```

[image is provided by kubernetes itself]

image name is *nginx-ingress-controller:*<some version>

[it is not a normal *nginx* image, it is made specific for kubernetes ingress]

- In the args, **--configmap=$(POD_NAMESPACE)/nginx-configuration** and POD_NAMESPACE is a env variable which is equal to *metadata.namespace*

  means, it is the namespace where *Deployment* of *nginx-ingress* is present.
- So it is just saying the nginx ingress controller to see the ConfigMap (Kubernetes resource) present inside the given namespace.

  Default/nginx-configuration (if deployment is present in default namespace)
- ```
  kind: ConfigMap
  apiVersion: v1
  metadata:
    name: nginx-configuration
  ```

  ConfigMap should look something like this. Name is ***nginx-configuration***
- So its like

  default/nginx-configuration

- Create a Service of type NodePort to expose the deployment

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-ingress
spec:
  type: NodePort
  ports:
  - port: 80
    targetPort: 80
    protocol: TCP
    name: http
  - port: 443
    targetPort: 443
    protocol: TCP
    name: https
  selector:
    name: nginx-ingress
```
```

- Ingress controller needs to have some specific permissions to do something in the cluster.

  So, it needs to have a **ServiceAccount** with specific permissions.

```yaml
apiVersion: v1
kind: ServiceAccount
metadata:
  name: nginx-ingress-serviceaccount
```

- Creating **Ingress Resource** (kind: Ingress)

  it specifies rules to route according to path or domain itself.

  - In case of a single service (*wear* for example in our case)

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ingress-wear
spec:
  defaultBackend:
    service:
      name: wear-service
      port: 80
```

    - defaultBackend means when no route rule is specified, this is the service where request is to be routed.
  - You can add as many rules as you want.

- Adding multiple rules to Ingress resource
  - Rule-1, path based routing i.e.

    my-online-store.com, my-online-store.com/wear, my-online-store.com/watch ..etc.

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: ingress-wear-watch
spec:
  rules:
  - http:
      paths:
      - path: /wear
        backend :
          service:
            name: wear-service
            port: 80
      - path: /watch
        backend:
          service:
            name: watch-service
            port: 80
```

- Rule-2, pased on domain names

  wear.my-online-store.com, www.my-online-store.com, watch.my-online-store.com ..etc.

```yaml
kind: Ingress
metadata:
  name: ingress-host-based
spec:
  rules:
  - host: wear.my-online-store.com
    http:
      paths:
      - path: /
        backend:
          serviceName: wear-service
          servicePort: 80
  - host: watch.my-online-store.com
    http:
      paths:
      - path: /
        backend:
          serviceName: watch-service
          servicePort: 80
```

- The difference between the 2 is:

  first one has only 1 rule, but multiple paths.

  Second one has 2 rules, each having only one path that is **/** means the domain itself. no further paths.

- NOTE:
  - **Rules**: one rule per host (or per domain)

    first one had only one rule because only one domain was there.

    Multiple paths (.../wear, .../watch)
  - **Paths**: multiple paths per rule (or per host)

    second one had 2 rules because 2 domains were there (wear..., watch...)
  -
-
- F

- How ingress class, ingress controller, ingress resources are linked?
  - IngressClass contains the controller type.

    nginx-ingress.

    Let say 2 different controllers are present.

    nginx-ingress and *traefik*
  - Now you need to create 2 **IngressClass** (each one pointing to one controller).
    - ```
      apiVersion: networking.k8s.io/v1
      kind: IngressClass
      metadata:
        name: nginx
      spec:
        controller: k8s.io/ingress-nginx
      ```
    - ```
      apiVersion: networking.k8s.io/v1
      kind: IngressClass
      metadata:
        name: traefik
      spec:
        controller: traefik.io/ingress-controller
      ```
    - Here, controller's values are hard-coded. Its not name of ingress controller's deployment or something else.

      For one type of controller, one hard-coded value is there.
  - Now, mention the IngressClass in

    IngressResource.*spec.ingressClassName*
    - ```
      apiVersion: networking.k8s.io/v1
      kind: Ingress
      metadata:
        name: nginx-app
      spec:
        ingressClassName: nginx
        rules:
      ```
    - ```
      apiVersion: networking.k8s.io/v1
      kind: Ingress
      metadata:
        name: traefik-app
      spec:
        ingressClassName: traefik
        rules:
      ```

- IngressClass will automatically selects the controller (deployed controller running in Pods) because both the controllers are of different types (nginx-ingress, traefik)
- But in case of same types of controllers, for example 2 *nginx-ingress* controllers are present then
  - IngressClass will be same only, because the IngressClass.*spec.controller* will contain hard-coded value only according to the type of controller.
    [here only one type of controller is there *nginx-ingress*]
- Create 2 IngressClass having different names, but same IngressClass.spec.controller

```yaml
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
  name: public-nginx
spec:
  controller: k8s.io/ingress-nginx
```

```yaml
apiVersion: networking.k8s.io/v1
kind: IngressClass
metadata:
  name: internal-nginx
spec:
  controller: k8s.io/ingress-nginx
```

[only *IngressClass.metadata.name* in both are different]

- Create 2 Ingress (ingress resource) giving Ingress.spec.ingressClassName as the IngressClass's names.

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: internal-app
spec:
  ingressClassName: internal-nginx
  rules:
```

```yaml
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: public-app
spec:
  ingressClassName: public-nginx
  rules:
```

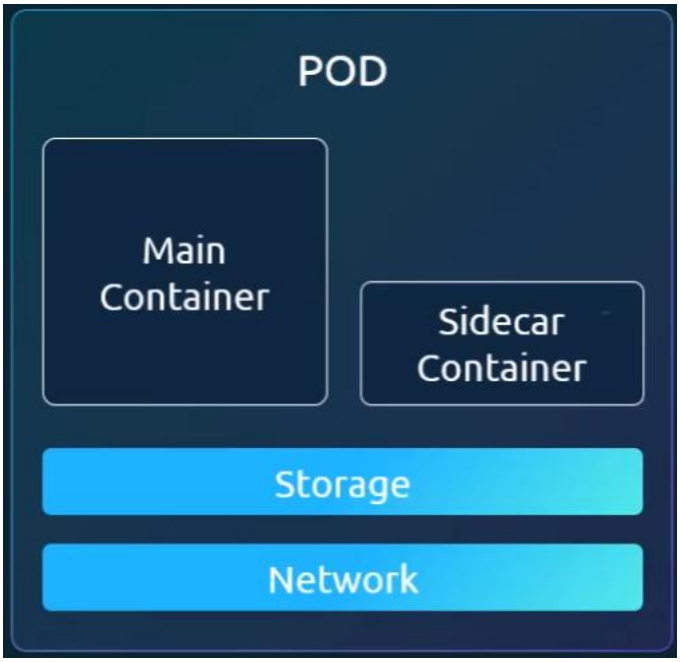
- Create 2 deployments of controllers (image of type *nginx-ingress* on both)
  -

```yaml
spec:
  containers:
    - name: nginx-ingress
      image: registry.k8s.io/ingress-nginx/controller:v1.12.0
      args:
        - /nginx-ingress-controller
        - --ingress-class=public-nginx
        - --controller-class=k8s.io/ingress-nginx
```

```
spec:
  containers:
    - name: nginx-ingress
      image: registry.k8s.io/ingress-nginx/controller:v1.12.0
      args:
        - /nginx-ingress-controller
        - --ingress-class=internal-nginx
        - --controller-class=k8s.io/ingress-nginx
```

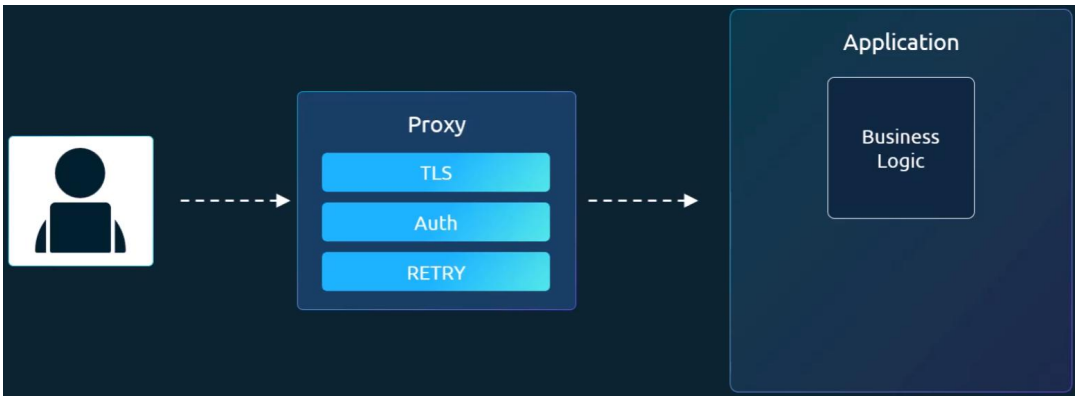
- Here we need **--controller-class=<**hard coded value of ingress controller**>**
- Also you can write these on ***ConfigMap*** as well just like the before.

Service Mesh

- **Sidecars**
  - 

  - One Pod can have multiple containers, one Main container and others sidecar containers.
  - Sidecars are responsible for Log shipping, Monitoring, File loading ..etc so that Main container can only focus on Business logic.
  -
```
containers:
 - name: nginx-container
   image: nginx
   volumeMounts:
   - name: shared-data
     mountPath: /usr/share/nginx/html
 - name: sidecar-container
   image:  fluent/fluentd
   volumeMounts:
   - name: shared-data
     mountPath: /pod-data
```

- **Envoy**
  - It is a proxy.
  - In a application, along with Business Logic, other things will be there like *TLS, Auth, RETRY* .
    These should be maintained in a nother layer called **Proxy** so that Developers will focus on writing Business Logic.

- Proxy receives the request anf forwards to the application.
- 

- **envoy** is a proxy or communication bus that runs as a Sidecar container.

- Microservices
  - When you convert a monolith to microservice, for all the services you need to handle the below things
    authentication, authorization, networking, logging, monitoring, tracing.
  - Networking part like rate limiting, circuit breakers, retry & timeouts are handled in application level.
  - Proxy handles all this so that developers will focus on writing business logic only.
  - Application (services) will call the other services, that cannot be changed because proxy doesn't know which service the request will be forwarded to.
    Without Proxy:
    service-1 --- Service of service-2(Kubernetes Service) **--- service-2**
    with Proxy:
    service-1 --- Proxy --- Service of service-2**(**Kubernetes Service) **--- service-2**

- **istio**
  - We don't add Sidecar container (*envoy*) by ourself, just install ***istio***, it automatically inject the Sidecar to each pod.
  - After running a deployment or pod, *istio* receives the PodSpec and update that so kubelet create the proxy container (sidecar) inside the pod.
  - It consists of ***dataplane*** (which contains the ***envoy*** only) and ***control plane*** (runs on master nodes).

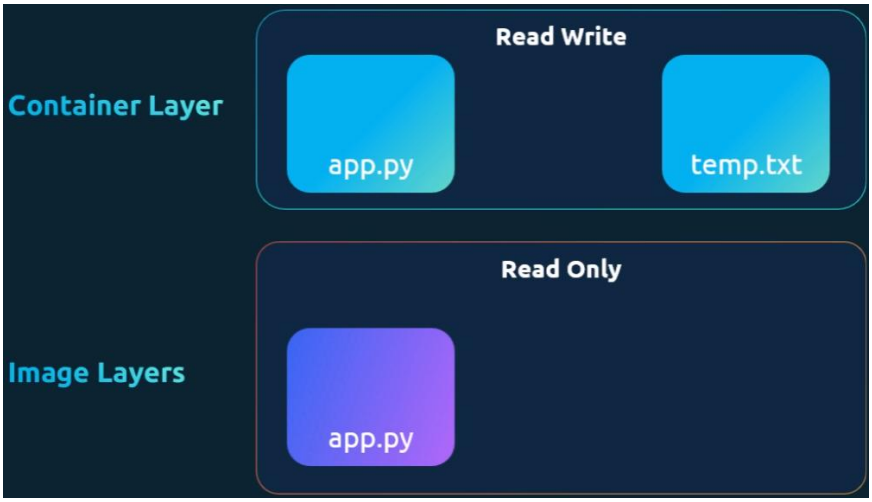
previously control plane was consisting of **3 daemons**: ***Citadel*** (certificate generation)***, Pilot***(service discovery)***, Galley***(validating configuration files)

- Currently, those 3 daemons combined to a single daemon called ***istiod*** (it's a Deployment)
- You need to *enable istio sidecar injection in the namespace* otherwise it'll not inject envoy proxy as sidecar on those namespace.

- In Mesh architecture, role of control plane: manages all traffic into and out of services via proxies.
- Istio agent delivers configuration secrets to Envoy proxies.
- In mesh architecture, proxies communicate with each other through **data plane**.
- Observability: logs, metrics, tracing
  Dynamic configuration: updating routing/config without restarting
  Manual TLS (mTLS): secure encrypted communication between services
  Service discovery: finding services automatically.
-
- F
-

Storage

- Docker image and container
  - Docker image consists of ***read-only*** layers. (Image Layers)
    layers are created based on the Dockerfile
  - When a container is created using the Docker image, a new layer is created on top of those image layers.
    This new layer is ***writable***. (Container Layer)
  - Container layer's life is same as the container.
    When container is destroyed, this container layer will also be destroyed.
  - Lets say in the image layer (in Dockerfile you do copy some files) a file was copied, so it'll be available inside all the containers created using that image.
    That file will be ***read-only*** and that is being shared among all the containers.
    You can edit that file, but when you'll save that, *Docker creates a copy of that file in **Read Write** layer (**Container Layer**) and whatever modification you'll do, that'll be there in the Container layer. Image layer will not be affected*.



- Docker **volume**
  - The data present inside the container will be destroyed once the container is destroyed.
    To persist those data, you can create Docker volume and attach that to the container.

- Create a volume using the command

  docker volume create <volume name>

  docker volume create data_volume
- Add the volume while running a container

  docker run -v <volume name>:<mount directory path> <docker image name>

  docker run -v data_volume:/var/lib/mysql mysql
- Docker volumes are present inside the directory

  /var/lib/docker/volumes

  in linux host.
- Even if you don't create a volume, but mention in *docker run* command, then Docker will automatically create a volume and attach it.
- This is called **Volume Mounting**.

- Bind mount
  - Mount a *directory* instead of a *docker volume*.
  - **docker run -v** <host directory path>:<container directory path> <docker image>

    both *volume* and *bind mount* has same command format.

    If it contains `./` or `../` or `/` then it'll take this as a directory, otherwise as a volume.

- Using **-v** flag to mount is old, nowadays **--mount** is used.
  - **docker run --mount type=**<mount type>**,source=**<volume name or host directory>**,target=**<container mount directory> **mysql**

    docker run --mount

    type=bind,source=/data/mysql,target=/var/lib/mysql mysql

    for volume mount, *type* need not to be provided. Docker by default takes the *type* as *volume* if not provided.

- **Storage Drivers**
  - It manages all the things like create a writable layer (container layer), copy files to that layer for modification, ..etc.
  - Selection of storage drivers depends upon the underlaying OS.
  - Ex: Ubuntu uses AUFS by default.
  - Docker chooses the best storage driver available based on the OS.

- **Volume Drivers**
  - *Volumes* are not handled by *Storage Drivers*.
  - Default volume driver is **local**
  - Other third party options are: Convoy, Flocker, NetApp, gcd-docker, RexRay ..etc
  - You can mention the volume driver while creating a container

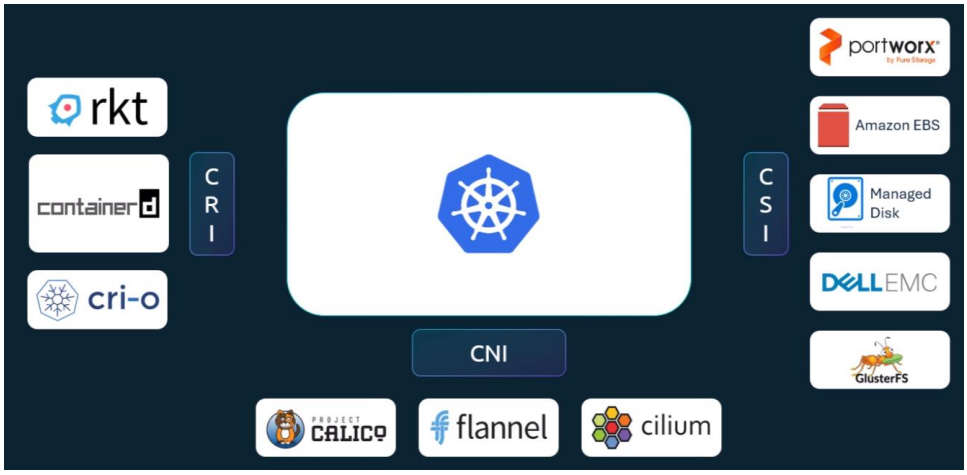
    docker run -it --name mysql **--volume-driver *resray/ebs* --mount src=**ebs-vol**,target=**/var/lib/mysql mysql

    it'll create a container and attach a *AWS* EBS volume to it.

CSI (Container Storage Interface)

- We saw **CRI** (Container Runtime Interface) [to support multiple container runtime, not only containerd], **CNI** (Container Network Interface) [any third party can create cni plugin according to CNI and can be used in Kubernetes].

  Similarly, **CSI** was developed to support multiple storage solutions.



  - ~~CSI is not a Kubernetes specific standards~~,

    Its universal.

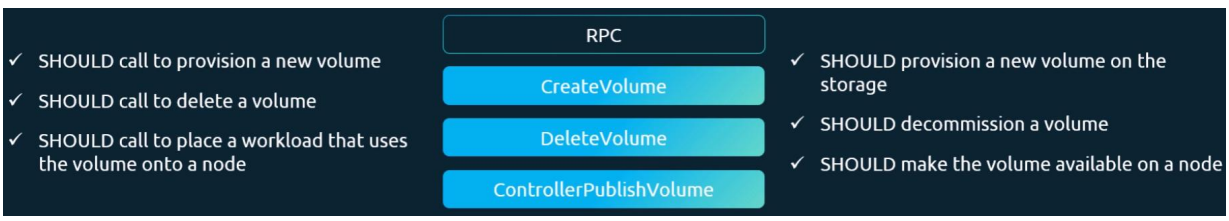
    So that any orchestration tool can use any kind of Storage solutions.
  - Kubernetes, Cloud Foundry, Mesos are on-board with CSI.

- CSI defines a series of RPCs used by **container orchestrators** to interact with **storage drivers**.

  [RPC: Remote Procedure Call

  One program calls a function/method running in another system/process over network or IPC, as if it were a local function.]

  - Volume creation:

    when pod is deployed and requires a storage solution, Kubernetes initiates “Create Volume” RPC call.
  - Same for Volume deletion as well.
  - 

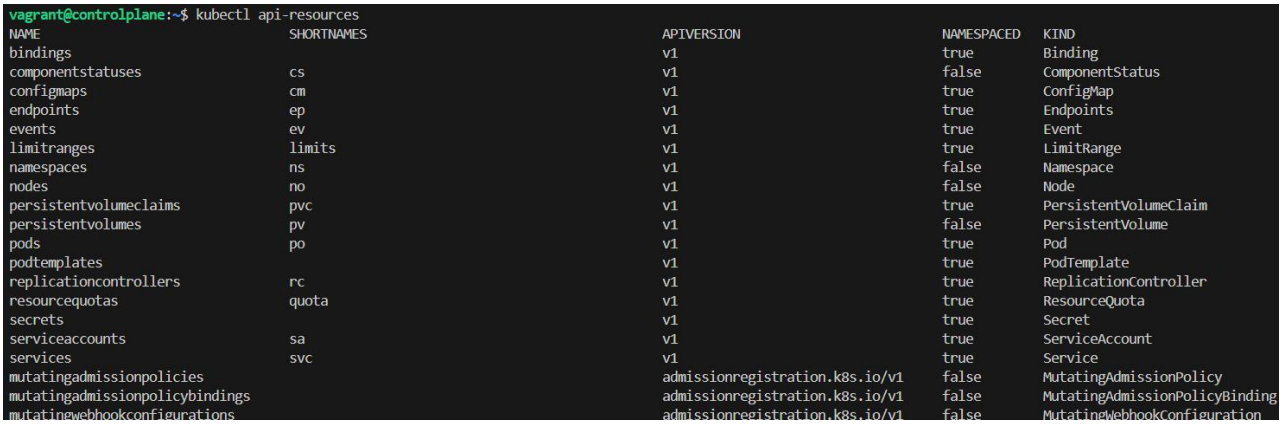
Volumes

々

々

Important Points

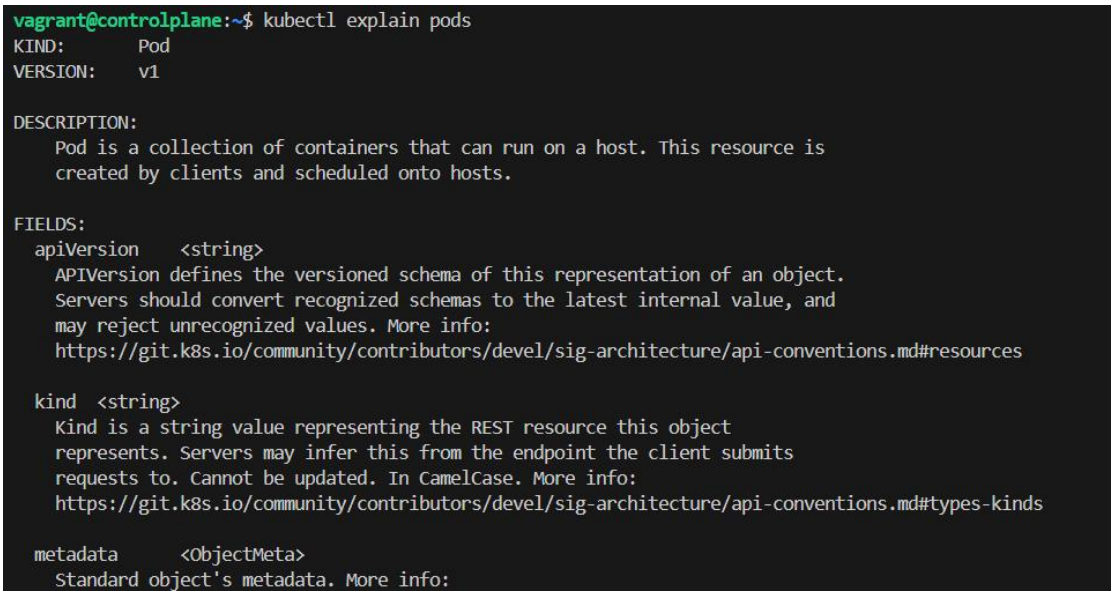
- Try to use *imperative commands* as much as you can in Exam.
- If you forget the keywords and resources, then execute ***kubectl api-resources*** you'll get the keywords.



```
vagrant@controlplane:~$ kubectl api-resources
NAME                              SHORTNAMES   APIVERSION                        NAMESPACED   KIND
bindings                                       v1                                true         Binding
componentstatuses                 cs           v1                                false        ComponentStatus
configmaps                        cm           v1                                true         ConfigMap
endpoints                         ep           v1                                true         Endpoints
events                            ev           v1                                true         Event
limitranges                       limits       v1                                true         LimitRange
namespaces                        ns           v1                                false        Namespace
nodes                             no           v1                                false        Node
persistentvolumeclaims            pvc          v1                                true         PersistentVolumeClaim
persistentvolumes                 pv           v1                                false        PersistentVolume
pods                              po           v1                                true         Pod
podtemplates                                   v1                                true         PodTemplate
replicationcontrollers            rc           v1                                true         ReplicationController
resourcequotas                    quota        v1                                true         ResourceQuota
secrets                                        v1                                true         Secret
serviceaccounts                   sa           v1                                true         ServiceAccount
services                          svc          v1                                true         Service
mutatingadmissionpolicies                      admissionregistration.k8s.io/v1   false        MutatingAdmissionPolicy
mutatingadmissionpolicybindings                admissionregistration.k8s.io/v1   false        MutatingAdmissionPolicyBinding
mutatingwebhookconfigurations                  admissionregistration.k8s.io/v1   false        MutatingWebhookConfiguration
```

[you can see there is a column 'KIND' which you can directly write in the yaml]

  - Shortnames are also there to use in *imperative commands* which will SAVE TIME.
  - If you want to get a documentation regarding any specific resource, you can use ***kubectl explain*** *<rource name>* command.



```
vagrant@controlplane:~$ kubectl explain pods
KIND:       Pod
VERSION:    v1

DESCRIPTION:
    Pod is a collection of containers that can run on a host. This resource is
    created by clients and scheduled onto hosts.

FIELDS:
  apiVersion    <string>
    APIVersion defines the versioned schema of this representation of an object.
    Servers should convert recognized schemas to the latest internal value, and
    may reject unrecognized values. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#resources

  kind  <string>
    Kind is a string value representing the REST resource this object
    represents. Servers may infer this from the endpoint the client submits
    requests to. Cannot be updated. In CamelCase. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-conventions.md#types-kinds

  metadata      <ObjectMeta>
    Standard object's metadata. More info:
```

  - To go deeper into the specific fields of that resources,

    kubectl explain *<resource name>.<field name>*

    [you can nest these any number times:

    resource.field.sub_field.sub_sub_field .... etc]

```
vagrant@controlplane:~$ kubectl explain pods.spec
KIND:       Pod
VERSION:    v1

FIELD: spec <PodSpec>


DESCRIPTION:
    Specification of the desired behavior of the pod. More info:
    https://git.k8s.io/community/contributors/devel/sig-architecture/api-convent
    PodSpec is a description of a pod.

FIELDS:
  activeDeadlineSeconds <integer>
    Optional duration in seconds the pod may be active on the node relative to
```

If you want, more comprehensive, then use

kubectl explain <resource name> --recursive

```
vagrant@controlplane:~$ kubectl explain pods --recursive
KIND:       Pod
VERSION:    v1

DESCRIPTION:
    Pod is a collection of containers that can run on a hos
    created by clients and scheduled onto hosts.

FIELDS:
  apiVersion    <string>
  kind  <string>
  metadata      <ObjectMeta>
    annotations <map[string]string>
    creationTimestamp   <string>
```

- To get a basic yaml file instead of writing from scratch, use ***--dry-run=client -o yaml*** with the command.

  [with *run, create* for static] [with *get* if the object is already running]

  kubectl create deployment my-dep --image=nginx --dry-run=client -o yaml

  kubectl run my-pod --image nginx --dry-run client -o yaml

```
vagrant@controlplane:~$ kubectl run my-pod --image=nginx --dry-run=client -o yaml
apiVersion: v1
kind: Pod
metadata:
  labels:
    run: my-pod
  name: my-pod
spec:
  containers:
  - image: nginx
    name: my-pod
    resources: {}
  dnsPolicy: ClusterFirst
  restartPolicy: Always
```

-
```
vagrant@controlplane:~$ kubectl create deployment my-dep --image=nginx --dry-run=client -o yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  labels:
    app: my-dep
  name: my-dep
spec:
  replicas: 1
  selector:
    matchLabels:
      app: my-dep
  strategy: {}
  template:
    metadata:
      labels:
        app: my-dep
    spec:
      containers:
      - image: nginx
        name: nginx
        resources: {}
```

- ***config*** file is present in ***~/.kube/config*** which contains all the details about *users, clusters, contexts*.
  - ***cluster*** contains the list of *kube-api-server endpoint* & *certificate* (for identity).
  - ***users*** contains the list of users each having a *unique name, key, certificate*.
  - ***context*** contains the combination of triplet i.e. *user, cluster, namespace*. [default namespace]
  - When you switch a context, the user will be updated, the cluster will be updated, and the default namespace will be updated.
    You can override the default namespace with the flag **--namespace=<**ns name**>**
  - ***kubectl config*** --help
    command you can use to see the usage of different fields of configs.
- TAINT, TOLERATIONS, NODE SELECTOR, NODE AFFINITY,
  - ***taint*** is to prevent a pod to be scheduled on a node (taint is applied on node).
  - ***toleration*** is to make a pod tolerant to get scheduled to a *taint* pod (toleration is applied on pod).

- ***nodeSelector*** is used to assign a pod to a node only if the selector labels matches with the node's labels. [HARD selection, if not matches then it is not scheduled].
- ***nodeAffinity*** is advanced version of *nodeSelector*, here *hard* and ***soft*** selection can be there with advanced selection queries.
- Sometimes you need to use both the things ***taint***, ***toleration*** and ***nodeAffinity*** to solve the real-world problem.

- You can give ***requests*** and ***limits*** on the pods.
  - ***requests*** means the pod's minimum requirement (cpu and memory) to be scheduled to a node.
  - ***limits*** means the pod can use that much amount of resources at max.
    - If ***cpu*** usage exceeds the *limit*, then it ***slows down***.
    - If ***memory*** usage exceeds, then pod is ***killed***.
  - Instead of giving this *requests* and *limits* in all the pods, you can create ***LimitRange*** (kind) which will contain the *request, limits, max, min* .
    - If any pod doesn't have the request and limit fields, then it'll inject its default values.
  - ***LimitRange*** works at ***namespace*** level.
- ***DaemonSet*** is same as ***ReplicaSet***, it just create a replica of the pod in each worker node.
- ***kubeadm*** creates static pods on control plane nodes to run the controlplane resources.
- API Server stores the details of all the pods in ***etcd*** (*normal pods* or *static pods*).

  in the annotation 2 extra fields are there in case of *static pods* which differentiate it from the normal pods.
  - When you edit a pod using *kubectl edit* command, it updates etcd and send *kubelet* (in streaming connection) the new *PodSpec* then *kubelet* compares it with the existing *PodSpec*.

    Then changes if allowed otherwise reject.

    In each changes, *containerd* creates new container and the container ID is changed.

- In case of *static pods*, you can't change using *kubectl edit* command in control plane nodes.
  You need to update the *yaml* file inside the *manifest* directory of the target node (where the static pod is running).

- *READ THE THEORY HAVING HEADING "Resources inside Pods vs Resources as normal Processes"*; DON'T SKIP IT.

-

- F

- F

Linux Specific

- **more** <fille path>

  read file in scrollable manner.

- **alias** <command> <alias> (in bash, not sh)

  alias date dt

- **$SHELL** : this variable contains current shell.

- **$PATH** : colon (:) separated paths, if its not there then command can't run.

  To add a path

  export PATH=$PATH:<*required path*>

- **chsh** : to change current shell

- **uname -r** : kernel current version

  <kernel version>:<major version>:<minor version>:<patch release>-<distro specific info>

  (output format)

- Memory management
  - Kernel space

    kernel and kernel related processes can only access this.

    They have access to hardware directly.

    Ex: device drivers
  - User space

    application/programs, make *system calls* to kernel to get hardware.

- Lets say you plug a pen-drive or hard-disk.

  Kernel space contains ***device drivers*** which detects this and emit ***uevents*** .

  User all *uevents* is sent to *User space*'s ***device manager*** daemon which is ***udev***.

  This *udev* dynamically creates a device (ex: /dev/sdb1) and attach it to the device.

- Commands
  - **dmesg** : display messages of a area of kernel called *ring buffer*. Also logs from hardware.
  - **lspci** : info about all *pci* devices configured in system.

- **lsblk** : list block devices

  disks & their partitions.

```
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
sda       8:0    0   100G  0 disk
├─sda1    8:1    0     1G  0 part /boot
├─sda2    8:2    0    50G  0 part /
└─sda3    8:3    0    49G  0 part /home
sdb       8:16   1    16G  0 disk
└─sdb1    8:17   1    16G  0 part /media/usb
```

  [sda: disk, sda1 to sda3 partitions]

- **lscpu** : info about CPU
- **lsmem** : info about memory
- **free -m** : total vs used memory
- **lshw** : details about the hardware

- Everything is file in linux

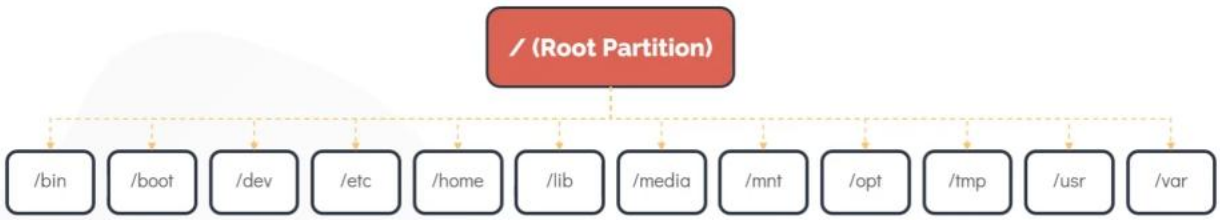
  3 types of files:

  - Regular file : image, scripts, configuration/data files
  - Directory : /home/username, /root, ..etc
  - Special files :

    it is of 5 types

    - Character files : devices to communicate with IO devices. Present inside **/dev** directory.
    - Block files : block devices (hard-disk ..etc). it is also present inside **/dev** directory.
    - Links : can be *hard links*, *soft links* .
    - Socket: enables communication between 2 processes.
    - Named Pipes:

- Filesystem hierarchy

  - 

  - **/opt** : third party programs installation directory.
  - **/mnt** : mount filesystems *temporarily*.

- **/tmp** : keep temporary data.
- **/media** : all external media are mounted in this.

  df -hp (displays all mounts)
- **/bin** : all commands like cp, mv ..etc present here.
- **/etc** : most of the configuration files are present.
- **/lib** : lib64

々